Sheet #	Content
01A.	Table of Content
01B.	Revision History
01C.	Block Diagram
01D.	GPIO Map
02.	PM8916 Control and MPP/Clock
03.	PM8916 Charging
04.	PM8916 GPIO/MPP
05.	PM8916 Buck converter
06.	PM8916 LDO circuits
07.	PM8916 CODEC
08.	MSM8916 Control
09.	MSM8916: EBI
10.	MSM8916: GPIO
11.	MSM8916: MIPI and RF Control
12.	MSM8916: POWER1
13.	MSM8916: POWER2
14.	MSM8916:GND
15.	MEMORY:LPDDR3+EMMC
16.	Battery Connector
17.	Subboard Connector
18.	Mic and Receiver
19.	EARPHONE
20.	LCD interface and backlight
21.	Main/Slave Camera and Flash
22.	Sensors
23.	SIM/TF card
24.	Keypad/LED/Status indicator
25.	Touch Interface
26.	Test Point/GND/Shields
27.	WTR1605L TX
28.	WTR1605L RX
29.	WTR1605L POWER
30.	WTR1605L POWER DISTRIBUTION
31.	LTE/W/TD/GSM Antenna Switch
32.	TRX_Front-High Band QFE2340
33.	TRX_Front-Low Band QFE2320
34.	RESERVED
35.	TRX_UMTS_B1/2/5/8
36.	PRX_B34/39/3
37.	PRX_LTE_HB
38.	DRX_Antenna Switch
39.	QFE1550 DRX Tunner
40.	B7/40/41 DRX Switch
41.	B39/B1/B3 DRX Switch
42.	WTR2605_SECONDARY PATH
43.	WTR2605_POWER DISTRIBUTION
44.	Antenna_SECONDARY PATH
45.	RESERVED
46.	RESERVED
47.	CDMA BC0(Voice) TRX
48.	ET_APT
49.	WCN3620
50.	WIFI FEM
51.	GPS/X0 DISTRIBUTOR
52.	NFC

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

Revision History

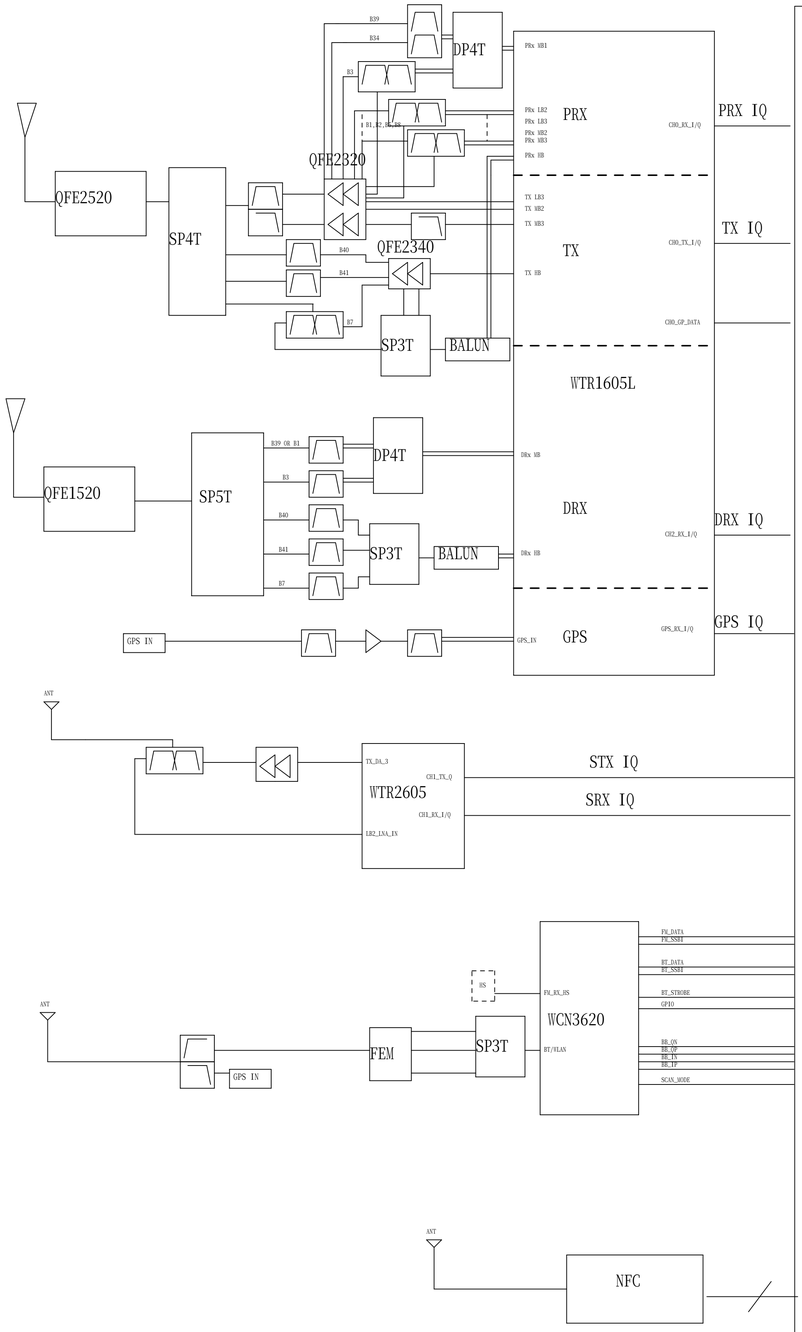
SCHEMATIC MCN	HW VERSION	VERSION	DESCRIPTION OF CHANGE
	EVT1.0	A	EVT1.0 Initial Release

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
SCHEMATIC, CCA, MAIN BOARD, QRD8916_2-4-2, CHCC CSFB, EVT

Size	Docet Name	Rev
Date:		Sheet of



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

MSM8916 GPIO Configuration For QRD8916

GPIO_0	NC	GPIO_41	WCSS_WLAN_DATA_1	GPIO_82	WTR1_SWITCH_CTL3
GPIO_1	NC	GPIO_42	WCSS_WLAN_DATA_0	GPIO_83	WTR1_GSM_PA_HB_EN
GPIO_2	NC	GPIO_43	WCSS_WLAN_SET	GPIO_84	WTR1_SWITCH_CTL0
GPIO_3	NC	GPIO_44	WCSS_WLAN_CLK	GPIO_85	DRX_MODE_SEL2
GPIO_4	UART_MSM_TX	GPIO_45	WCSS_FM_SSBI	GPIO_86	NC
GPIO_5	UART_MSM_RX	GPIO_46	WCSS_FM_SDI	GPIO_87	WTR0_PA_R1
GPIO_6	SENSOR_I2C_SDA	GPIO_47	WCSS_BT_CTRL	GPIO_88	WTR1_PA_R1
GPIO_7	SENSOR_I2C_SCL	GPIO_48	WCSS_BT_DAT_STB	GPIO_89	WTR1_PA_R0
GPIO_8	NC	GPIO_49	LCD_ID	GPIO_90	DRX_MODE_SEL0
GPIO_9	NC	GPIO_50	CTP_ID	GPIO_91	GRFC_17
GPIO_10	NC	GPIO_51	MCAM_ID	GPIO_92	WTR1_PA_BCO_EN
GPIO_11	NC	GPIO_52	NC	GPIO_93	DRX_MODE_SEL1
GPIO_12	TP_RST_N	GPIO_53	UIM2_DATA	GPIO_94	WTR0_PA_R0
GPIO_13	TP_INT_N	GPIO_54	UIM2_CLK	GPIO_95	WTR1_SWITCH_CTRL2
GPIO_14	BUCK_SMB_I2C_SDA	GPIO_55	UIM2_RESET	GPIO_96	EXT_GPS_LNA_EN
GPIO_15	BUCK_SMB_I2C_SCL	GPIO_56	NC	GPIO_97	NC
GPIO_16	NC	GPIO_57	UIM1_DATA	GPIO_98	NC
GPIO_17	NC	GPIO_58	UIM1_CLK	GPIO_99	CH0_GSM_TX_PHASE_D0
GPIO_18	TP_I2C_SDA	GPIO_59	UIM1_RESET	GPIO_100	CH0_GSM_TX_PHASE_D1
GPIO_19	TP_I2C_SCL	GPIO_60	NC	GPIO_101	CH1_GSM_TX_PHASE_D0
GPIO_20	NFC_DISABLE	GPIO_61	UIM_BATT_ALM	GPIO_102	CH1_GSM_TX_PHASE_D1
GPIO_21	NFC_INT_N	GPIO_62	SMB_INT	GPIO_103	CH0_SSBI_WTR_RX
GPIO_22	NFC_I2C_SDA	GPIO_63	CDC_PDM_CLK	GPIO_104	CH0_SSBI_WTR_TX
GPIO_23	NFC_I2C_SCL	GPIO_64	CDC_PDM_SYNC	GPIO_105	CH1_SSBI_WTR_RX
GPIO_24	LCD_TE0	GPIO_65	CDC_PDM_TX	GPIO_106	CH1_SSBI_WTR_TX
GPIO_25	LCD_RST_N	GPIO_66	CDC_PDM_RX0	GPIO_107	KEY_VOL_UP_N
GPIO_26	CAM0_MCLK	GPIO_67	CDC_PDM_RX1	GPIO_108	NC
GPIO_27	CAM1_MCLK	GPIO_68	CDC_PDM_RX2	GPIO_109	NC
GPIO_28	SCAM_RST_N	GPIO_69	NC	GPIO_110	USB_HS_ID
GPIO_29	CAM_I2C_SDA	GPIO_70	RFFE1_CLK	GPIO_111	NC
GPIO_30	CAM_I2C_SCL	GPIO_71	RFFE1_DATA	GPIO_112	ACCL_INT1_N
GPIO_31	CAM_FLASH_TORCH_EN	GPIO_72	RFFE2_CLK	GPIO_113	ALSP_INT_N
GPIO_32	CAM_FLASH_TORCH_MODE	GPIO_73	RFFE2_DATA	GPIO_114	ACCL_INT2_N
GPIO_33	SCAM_PWDN	GPIO_74	NC	GPIO_115	GYRO_INT_N
GPIO_34	MCAM_PWDN	GPIO_75	GRFC_1	GPIO_116	NC
GPIO_35	MCAM_RST_N	GPIO_76	GRFC_2	GPIO_117	NC
GPIO_36	NC	GPIO_77	WTR1_GSM_PA_LB_EN	GPIO_118	NC
GPIO_37	FORCE_USB_BOOT	GPIO_78	TDS_B34/B39_PA_ON	GPIO_119	CTP_LED_EN
GPIO_38	NC	GPIO_79	WTR1_SWITCH_CTL1	GPIO_120	CAM_VCM_PWDN
GPIO_39	WCSS_BT_SSBI	GPIO_80	WTR0_RX_ON	GPIO_121	NC
GPIO_40	WCSS_WLAN_DATA_2	GPIO_81	WTR0_RF_ON		

PM8916 GPIO/MPP Configuration For QRD8916

GPIO_1	UIM_BATT_ALM	MPP_1	VDD_PX_BIAS_MPP
GPIO_2	NFC_CLK_REQ	MPP_2	SCAM_ID
GPIO_3	NC	MPP_3	VREF_DAC_MPP
GPIO_4	NC	MPP_4	PWM_OUT

Sensors I2C Address List For QRD8916

I2C GPIO	Device	MPN	Write Address	Read Address
GPIO[6:7]	Accelerometer	BMA223	0x30	0x31
	Magnetometer	MMC34160PJ	0x60	0x61
	Gyroscope	MPU-6050C	0xD0	0xD1
	ALS/Proximity	TMD27723	0x72	0x73
GPIO[18:19]	Touch Screen	FT6346	0x70	0x71
GPIO[29:30]	Front Camera	OV2680	0x20	0x21
	Rear Camera	OV5648	0x6C	0x6D

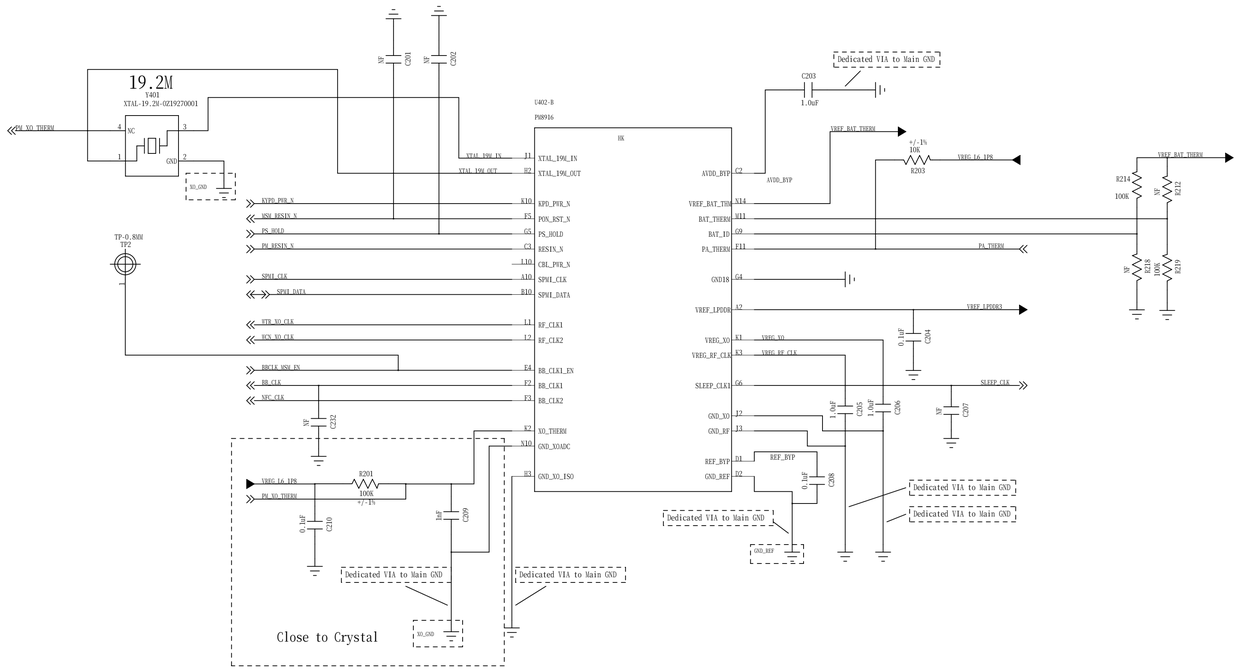
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
SCHEMATIC, CCA, MAIN BOARD, QRD8916,2-4-2, CHCC CSFB, EVT

Size | Sheet Name | Rev

Date: | Sheet of



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

Pins	Hi Z	GND
OPT_1	No SMB1360	SMB1360 present
OPT_2	No Ext APC buck (S5)	Ext APC buck (S5) present

PM8916 support +15V input protected

Dedicated sense trace to battery connector

Already have 47k resistor in subboard

Close to connector
if using PMIC's linear charger
C0301 should assemble 4.7uF/16V capacitor
R0307 should assebble 0 ohm
R0302 should NC
R0312 should be 0ohm
BAT_THERM switch to PM8916

SMB1360 can support +28V input protected

Software Caution:
Need software to config MSM
GPIO internal pullup

Dedicated differential sense trace from resistor

Dedicated differential sense trace to battery connector

NOTE1: Seperate the AGND from the PGND

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

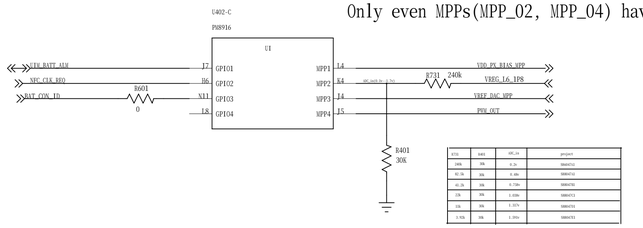
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

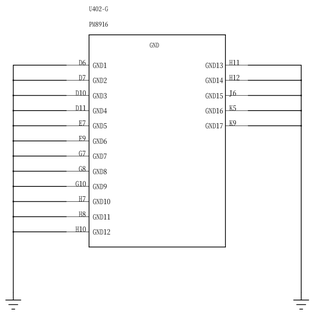
Title

Size | Sheet Name | Rev

Date: | Sheet of

Only odd MPPs (MPP_01, MPP_03) can be configured as analog outputs.
Only even MPPs(MPP_02, MPP_04) have current sink capability.





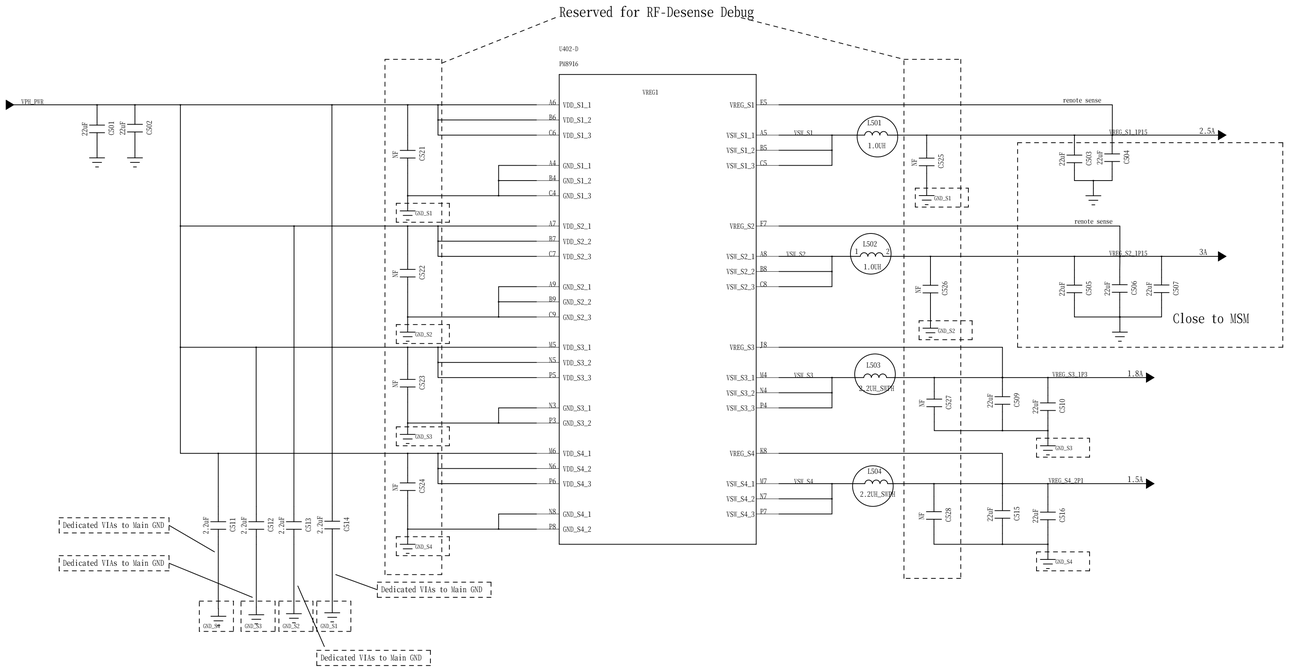
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



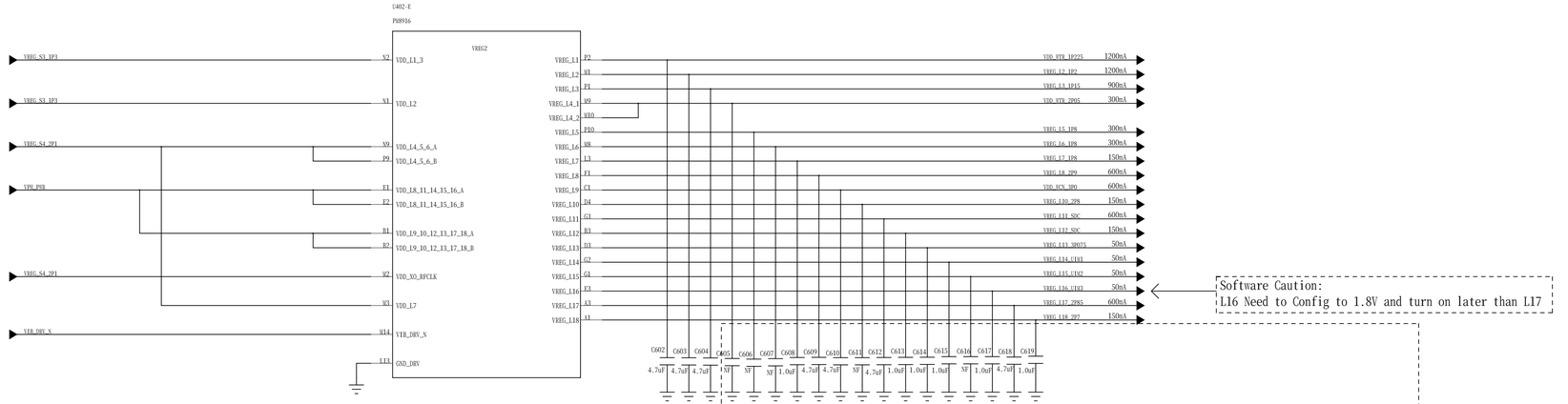
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



Software Caution:
L16 Need to Config to 1.8V and turn on later than L17

PSEUDO CAPLESS LDOs
L4, L5, L6, L7, L8, L9, L10, L11, L12, L13, L14, L15, L16, L17, L18

Local output capacitor can NC/Remove depends on remote load capacitors and PCB layout
LDOs L8, L9, L11, L17 each shall have total effective output capacitance of 4.7uF.
Please refer to PM8916 training slides(80-NK808-21) for limits on LDO output trace resistance and inductance
Rest of the LDOs shall each have total effective output capacitance of 1.0uF.

The Pseudo capless LDOs must comply to following tolerable routing requirement:
P50: R<400mOhms, L<45nH
P150: R<200mOhms, L<20nH
P300: R<150mOhms, L<18nH
P600: R<150mOhms, L<18nH

Please ensure when you DNI the output capacitor of Pseudo capless LDOs
the effective total capacitance at load side shall match above requirement

NC first

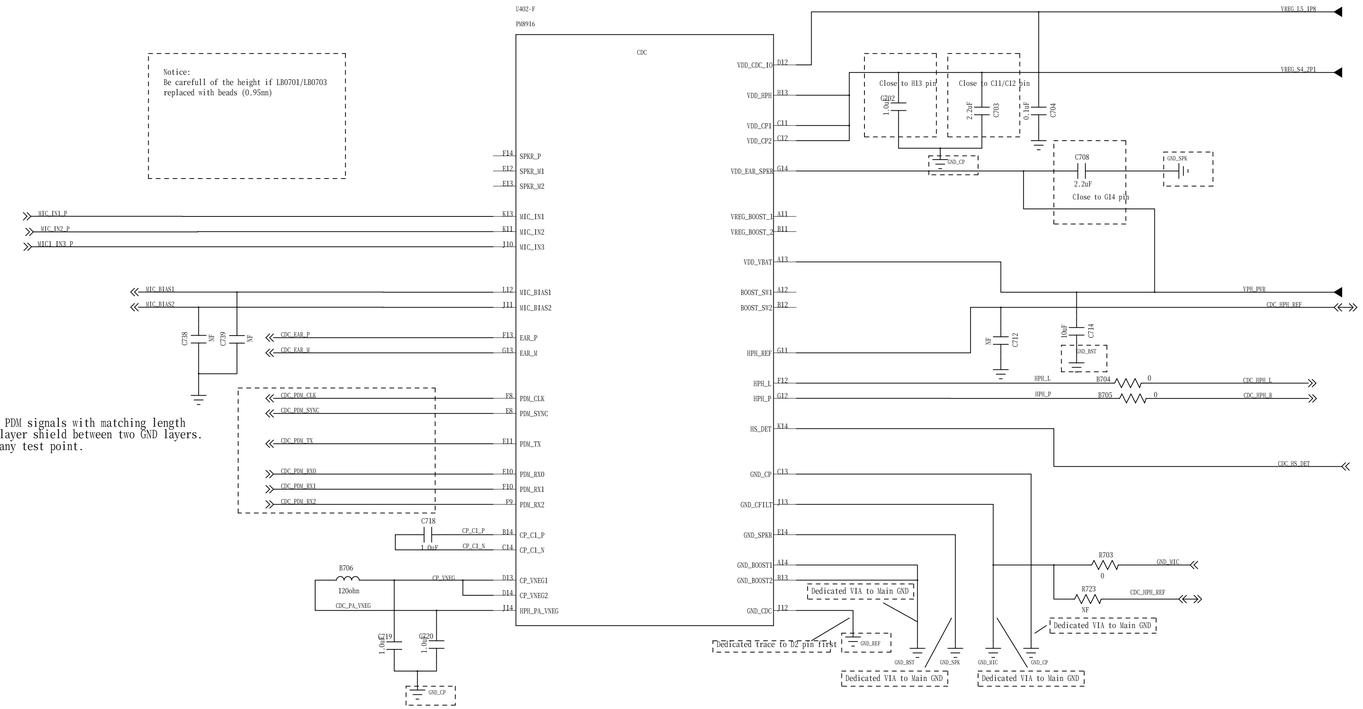
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

For BBCLK1,AD35 will be the first load and AL34 will be the final load
Should be a daisy chain route

U401-A

MSM8916

CONTROL

	AR8	VLAN_XO
BB_CLK	AL34	CXO
BBCLK_MSM_EN	A36	CXO_EN
SLEEP_CLK	C36	SLEEP_CLK
MSM_RESETX_N	F35	RESET_N
MSM_RESOUT_N	AY1	RESOUT_N
	AU32	MODE_0
	AT31	MODE_1
	E36	PS_HOLD
JTAG_RESOUT_N	K1	SRST_N
JTAG_TCK	H1	TCK
JTAG_TDI	H3	TDI
JTAG_TDO	J2	TDO
JTAG_TMS	L2	TMS
JTAG_TRST_N	K3	TRST_N
VREF_LPDDR3	J12	EBI_VREF_CA
	H21	EBI_VREF_D0_D2
VREF_LPDDR3	J26	EBI_VREF_D1_D3
	AN8	DNC1
	BA36	DNC2
	AY35	DNC3

TP8
TP-0.80H

TP26
TP-0.80H

Star routing from C0803 to PINs
NC others CAPs first

VREF_LPDDR3

NF C0801

NF C0802

0.1uF C0803

SDC1_CLK	AE36	SDC1_CLK
SDC1_CMD	AH35	SDC1_CMD
SDC1_DATA_0	AD37	SDC1_DATA_0
SDC1_DATA_1	AF38	SDC1_DATA_1
SDC1_DATA_2	AG34	SDC1_DATA_2
SDC1_DATA_3	AK37	SDC1_DATA_3
SDC1_DATA_4	AG40	SDC1_DATA_4
SDC1_DATA_5	AG38	SDC1_DATA_5
SDC1_DATA_6	AF39	SDC1_DATA_6
SDC1_DATA_7	AF35	SDC1_DATA_7
SDC2_CLK	R4	SDC2_SDCARD_CLK
SDC2_CMD	N6	SDC2_SDCARD_CMD
SDC2_DATA_0	F3	SDC2_SDCARD_D0
SDC2_DATA_1	R6	SDC2_SDCARD_D1
SDC2_DATA_2	T7	SDC2_SDCARD_D2
SDC2_DATA_3	P7	SDC2_SDCARD_D3

R802
0

R804
0

Close to MSM

USB_HS_DP	AC40	USB_HS_D_P
USB_HS_DM	AR39	USB_HS_D_M
USB_HS_ID	AA40	
USB_HS_REXT	AC38	USB_HS_REXT
USB_HS_SYSCLOCK	AU35	BB_CLK

R803
200
+/-1%

PMIC_SPMI_CLK	AK37	SPMI_CLK
PMIC_SPMI_DATA	AJ36	SPMI_DATA

C804 NF

C805 NF

SPMI Data trace length should be within +/-2mm of SMPI Clock trace length. 33pF shunt caps must be no more than 2mm away from the MSM.

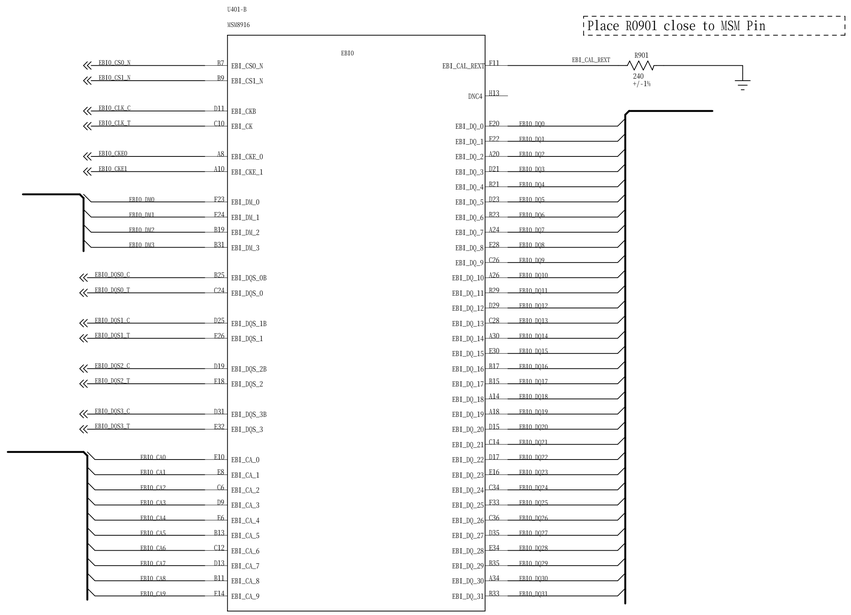
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

U401-C

MSM8916

GPIO1

BAT_ID_1V8	BA38 GPIO_0	GPIO_37 F4	FORCE_USB_BOOT
	BR39 GPIO_1 *	GPIO_38 R37	
	AY35 GPIO_2	GPIO_39 BC2	WCNSS_BT_SSBI
	AT37 GPIO_3	GPIO_40 BB5	WCNSS_HLAN_DATA_2
UART_WSH_TX	AT9 GPIO_4	GPIO_41 BB1	WCNSS_HLAN_DATA_1
UART_WSH_RX	AY1 GPIO_5 *	GPIO_42 BD3	WCNSS_HLAN_DATA_0
SENSOR_I2C_SDA	AY3 GPIO_6	GPIO_43 BA6	WCNSS_HLAN_SET
SENSOR_I2C_SCL	AY5 GPIO_7	GPIO_44 BB3	WCNSS_HLAN_CLK
LCD_2PWS_EN	BA5 GPIO_8	GPIO_45 AY5	WCNSS_FM_SSBI
	G2 GPIO_9 *	GPIO_46 AY9	WCNSS_FM_SDI
	B1 GPIO_10	GPIO_47 BA4	WCNSS_BT_CTRL
	E2 GPIO_11 *	GPIO_48 AY8	WCNSS_BT_DAT_STR
TP_RST_N	AU39 GPIO_12 *	GPIO_49 V37	LCD_ID
TP_INT_N	AU35 GPIO_13 *	GPIO_50 AA34	CTP_ID
RCCK_SMB_I2C_SDA	AU40 GPIO_14	GPIO_51 V35	WCAM_ID
RCCK_SMB_I2C_SCL	AV36 GPIO_15	GPIO_52 AA38	
	K7 GPIO_16	GPIO_53 L40	UIF2_DATA
	G6 GPIO_17	GPIO_54 K39	UIF2_CLK
TP_I2C_SDA	J6 GPIO_18	GPIO_55 U39	UIF2_RESET
TP_I2C_SCL	F4 GPIO_19	GPIO_56 AA36	
NFC_DISABLE	AT7 GPIO_20 *	GPIO_57 G40	UIF1_DATA
NFC_INT_N	AT6 GPIO_21 *	GPIO_58 J40	UIF1_CLK
FED_I2C_SDA	AU7 GPIO_22	GPIO_59 L38	UIF1_RESET
LED_I2C_SCL	BA2 GPIO_23	GPIO_60 V39	
LCD_TE0	AT5 GPIO_24	GPIO_61 AL38	UIM_BATT_ALM
LCD_RST_N	AU4 GPIO_25 *	GPIO_62 G34	I_SUB_IN1
CAM0_MCLK	B3 GPIO_26	GPIO_63 AJ40	CDC_PDM_CLK
CAM1_MCLK	C2 GPIO_27	GPIO_64 AR39	CDC_PDM_SYNC
SCAM_RST_N	F1 GPIO_28 *	GPIO_65 AK39	CDC_PDM_TX
CAM_I2C_SDA	B3 GPIO_29	GPIO_66 AJ38	CDC_PDM_RX0
CAM_I2C_SCL	F3 GPIO_30	GPIO_67 AK35	CDC_PDM_RX1
CAM_FLASH_TORCH_EN	D3 GPIO_31 *	GPIO_68 AL40	CDC_PDM_RX2
CAM_FLASH_TORCH_MODE	D1 GPIO_32	GPIO_69 L36	
SCAM_PWDN	G4 GPIO_33	GPIO_70 AN38	RFFE1_CLK
MCAM_PWDN	F5 GPIO_34 *	GPIO_71 AR38	RFFE1_DATA
MCAM_RST_N	A4 GPIO_35 *	GPIO_72 AT39	RFFE2_CLK
	C4 GPIO_36 *	GPIO_73 AU40	RFFE2_DATA

P3 P4 P5 P6

R1009 0 CAM0_MCLK_BB

R1010 0 CAM1_MCLK_BB

For RF De-sense Debug

C806 33PF

C807 33PF

Need software to config internal pullup

U401-D

MSM8916

GPIO2

LTE_BPI_OUT74	GRFC0 AP37	GPIO_74	
GRFC_1	GRFC1 AV39	GPIO_75	
GRFC_2	GRFC2 AP35	GPIO_76	
	GRFC3 BC4	GPIO_77	
TDS_B34/B39_PA_ON	GRFC4 BE4	GPIO_78	
LTE_BPI_OUT79	GRFC5 BC6	GPIO_79	
WTR0_RX_ON	GRFC6 BG5	GPIO_80	
WTR0_RF_ON	GRFC7 BG7	GPIO_81	
LTE_BPI_OUT82	GRFC8 BC38	GPIO_82	
	GRFC9 BC40	GPIO_83	
LTE_BPI_OUT84	GRFC10 BC8	GPIO_84	
B39_B3_SHITCH_SEL	GRFC11 BE6	GPIO_85	
	GRFC12 BD39	GPIO_86	
LTE_BPI_OUT87	GRFC13 BR37	GPIO_87	
LTE_BPI_OUT88	GRFC14 BR35	GPIO_88	
	GRFC15 BD35	GPIO_89	
DRX_MODE_SEL0	GRFC16 BA32	GPIO_90	
	GRFC17 BE38	GPIO_91	
LTE_B7_PA_EN	GRFC18 AY33	GPIO_92	
DRX_MODE_SEL1	GRFC19 BD33	GPIO_93	
WTR0_PA_R0	GRFC20 BC34	GPIO_94	
DRX_MODE_SEL2	GRFC21 BR31	GPIO_95	
EXT_GPS_LNA_EN	BC32	GPIO_96	
	C38	GPIO_97 *	
	A38	GPIO_98 *	
CH0_GSM_TX_PHASE_D0	BF34	GPIO_99	
CH0_GSM_TX_PHASE_D1	BE36	GPIO_100	
CH1_GSM_TX_PHASE_D0	BR33	GPIO_101	
CH1_GSM_TX_PHASE_D1	BF32	GPIO_102	
CH0_SSBI1_WTR_RX	AT35	GPIO_103	
CH0_SSBI1_WTR_TX	AV40	GPIO_104	
CH1_SSBI1_WTR_RX	AU36	GPIO_105	
CH1_SSBI1_WTR_TX	AV39	GPIO_106	
KEY_VOL_UP_N	H33	GPIO_107 *	
	K35	GPIO_108 *	
	J34	GPIO_109 *	

	B39	GPIO_110	USB_HS_ID
	C40	GPIO_111	GPIO_111 will be in O/P mode on Power On Reset, After Power On Reset, SW can configure it
	AE36	GPIO_112	I2S_2_D1
	D39	GPIO_113	ALSP_INT_N
	E40	GPIO_114	ACC1_INT2_N
	E38	GPIO_115	GYRO_INT_N
	AR38	GPIO_116	
	AV37	GPIO_117	I2S_2_WS
	AE36	GPIO_118	I2S_2_SCLK
	B40	GPIO_119	I2S_2_D0
	E39	GPIO_120	CAM_VCM_PWDN
	G38	GPIO_121	

P3

VREG_L5_1P8

R1024 10K

Internal Use.

SENSORS

VREG_L6_1P8

R1011 2.2K

R1012 2.2K

SENSOR_I2C_SDA

SENSOR_I2C_SCL

CAMERAS

VREG_L6_1P8

R1013 2.2K

R1014 2.2K

CAM_I2C_SDA

CAM_I2C_SCL

EXT BUCK, SMB

VREG_L5_1P8

R1015 2.2K

R1016 2.2K

RCCK_SMB_I2C_SDA

RCCK_SMB_I2C_SCL

TOUCHSCREEN

VREG_L6_1P8

R1017 2.2K

R1018 2.2K

TP_I2C_SDA

TP_I2C_SCL

LED

VREG_L5_1P8

R1019 2.2K

R1020 2.2K

LED_I2C_SDA

LED_I2C_SCL

I2C PULL-UP RESISTORS

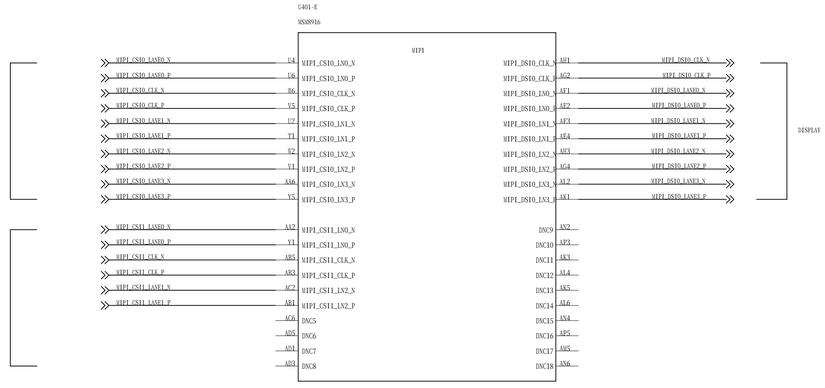
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

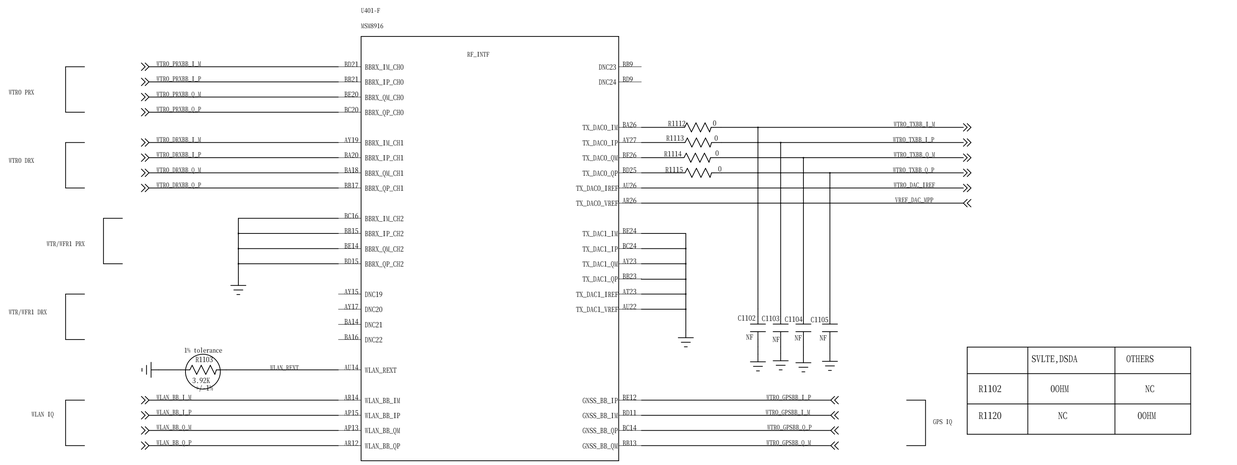
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of





	SVLTE,DSDA	OTHERS
R1102	0OHM	NC
R1120	NC	0OHM

These NC resistor is for SVLTE or DSDA. Change then to 0ohm on other modes.

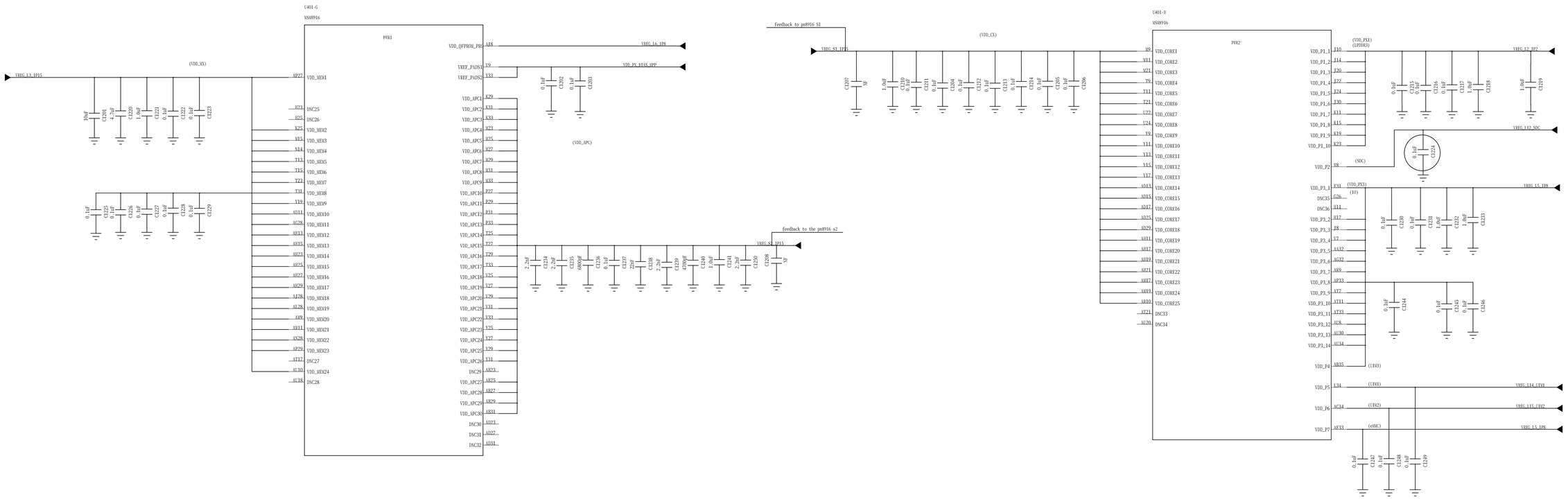
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



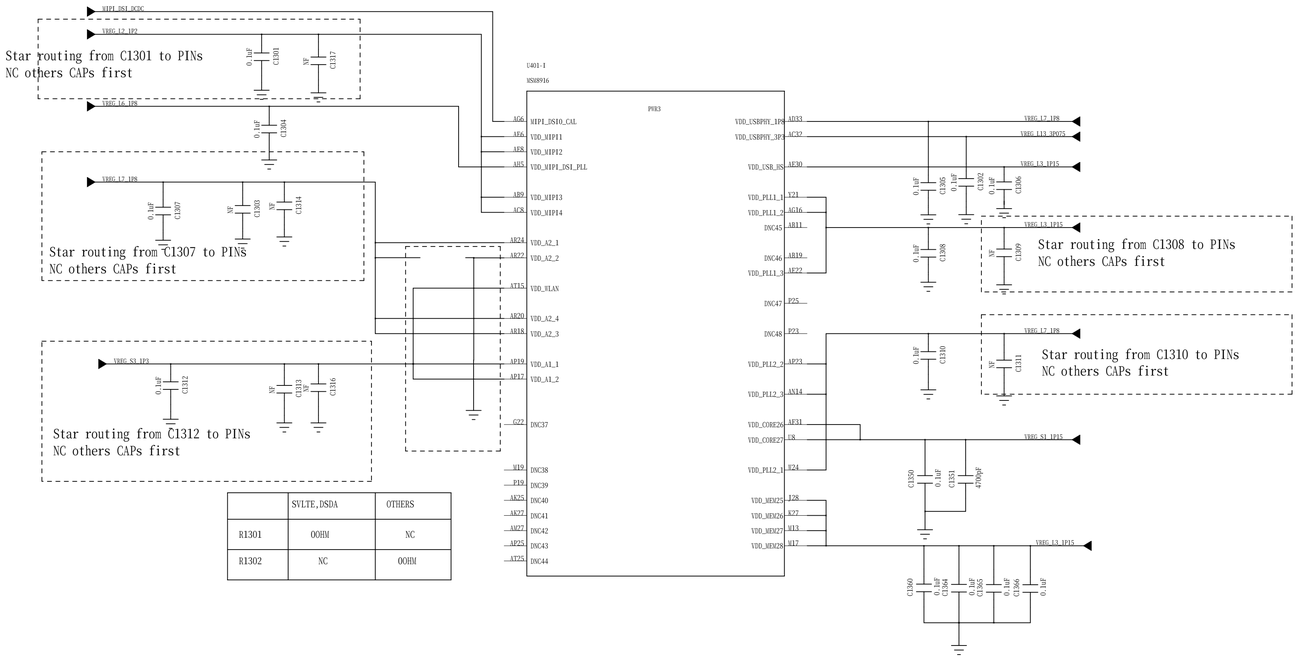
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



	SVLTE,DSDA	OTHERS
R1301	0OHM	NC
R1302	NC	0OHM

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

U401-K

MSM8916

GND2

V35	GND101	GND139	AF21
V37	GND102	GND140	AF23
V39	GND103	GND141	AF29
R16	GND104	GND142	AH31
R26	GND105	GND143	AH33
R28	GND106	GND144	AL30
R30	GND107	GND145	AP25
R32	GND108	GND146	AP29
R34	GND109	GND147	AP31
R36	GND110	GND148	AP33
R38	GND111	GND149	AP9
R40	GND112	GND150	AR4
AA24	GND113	GND151	AR6
AA26	GND114	DNC45	AT1
AA28	GND115	DNC46	AT3
AA30	GND116	GND154	AT29
AB15	GND117	DNC47	AU2
AB17	GND118	GND156	AU6
AP9	GND119	GND157	AU12
AF11	GND120	GND158	AU38
AF13	GND121	GND159	AV5
AF15	GND122	GND160	AV9
AF17	GND123	GND161	AV11
AF19	GND124	GND162	AV17
AF21	GND125	GND163	AV19
AF23	GND126	GND164	AV33
AF25	GND127	GND165	AY2
AF27	GND128	GND166	AY4
AF29	GND129	GND167	AY10
AG20	GND130	GND168	AY34
AH9	GND131	GND169	AY11
AH31	GND132	GND170	BA8
AJ34	GND133	GND171	BB7
AK11	GND134	GND172	BC36
AK13	GND135	GND173	BD37
AK15	GND136	GND174	BD8
AK17	GND137	GND175	BE10
AK19	GND138		

U401-J

MSM8916

GND1

A6	GND1	GND51	AC37
A12	GND2	GND52	AC39
A16	GND3	GND53	AC2
A22	GND4	GND54	AC4
A28	GND5	GND55	AC8
A32	GND6	GND56	AC27
B5	GND7	GND57	AC26
C8	GND8	GND58	AC28
C18	GND9	GND59	AC30
C20	GND10	GND60	AC32
C32	GND11	GND61	AC34
D5	GND12	GND62	AC36
D7	GND13	GND63	AC38
D27	GND14	GND64	AC40
E12	GND15	GND65	AD5
E7	GND16	GND66	AD9
F19	GND17	GND67	AD11
F25	GND18	GND68	AD13
F27	GND19	GND69	AD15
F29	GND20	GND70	AD17
F37	GND21	GND71	AD35
G8	GND22	GND72	AD37
G10	GND23	GND73	AD39
G12	GND24	GND74	AE8
G14	GND25	GND75	AE26
G16	GND26	GND76	AE28
G18	GND27	GND77	AE30
G28	GND28	GND78	AE32
G30	GND29	GND79	AE34
G32	GND30	GND80	AE36
H35	GND31	GND81	AE38
H37	GND32	GND82	AE40
J16	GND33	GND83	AT17
J18	GND34	GND84	AT19
J32	GND35	GND85	AT35
J36	GND36	GND86	AT37
J38	GND37	GND87	AT39
K5	GND38	GND88	AU26
K9	GND39	GND89	AU28
K17	GND40	GND90	AU30
K21	GND41	GND91	AU32
K37	GND42	GND92	AU34
L4	GND43	GND93	AU36
L6	GND44	GND94	AU38
L8	GND45	GND95	AU40
L30	GND46	GND96	AV11
L32	GND47	GND97	AV13
M5	GND48	GND98	AV15
M7	GND49	GND99	AV17
M35	GND50	GND100	AV19

U401-L

MSM8916

GND3

AJ4	GND176	GND216	AC30
AF5	GND177	DNC48	F21
AU7	GND178	GND218	G24
AJ2	GND179	GND219	G20
AK7	GND180	GND220	AN20
AT8	GND181	GND221	AB13
AU1	GND182	GND222	Y23
AW3	GND183	GND223	Y21
T3	GND184	GND224	Y23
T5	GND185	GND225	Y22
V3	GND186	GND226	AG18
W4	GND187	GND227	AF16
Y3	GND188	GND228	AR21
AA4	GND189	GND229	AD19
AA8	GND190	GND230	AF20
AC4	GND191	GND231	AD21
AD7	GND192	GND232	AJ24
AT27	GND193	GND233	AJ24
AU28	GND194	GND234	F21
BA28	GND195	GND235	E22
BB27	GND196	GND236	AW23
BC26	GND197	GND237	AV21
BC28	GND198	GND238	AU15
BD27	GND199	GND239	AU13
BE28	GND200	GND240	AV15
BR25	GND201	GND241	AV14
BA22	GND202	GND242	AY13
BC22	GND203	GND243	BA12
BD23	GND204	GND244	BC12
BF22	GND205	GND245	BD13
AT13	GND206		
AV13	GND207		
AE12	GND208		
AY21	GND209		
BD19	GND210		
BE18	GND211		
BB19	GND212		
BC18	GND213		
BD17	GND214		
BE16	GND215		

MIPI_DSI_LDO is configured to be LDO mode defaultly.
if using DC/DC mode,
L1401 need to be replaced with 2.2uH inductor

MIPI_DSI_DCDC

L1401
0_NF

C1401
2.2uF

CAD NOTE: Loop to be shorted with MSM AB7 pin

U401-H

MSM8916

NC_DNC

A2	NC1	NC24	AY27
A40	NC2	NC25	AY29
B1	NC3	NC26	AY31
D33	NC4	NC27	AY16
D37	NC5	NC28	AY18
F15	NC6	NC29	AY20
F17	NC7	NC30	AY22
H7	NC8	NC31	AY24
H15	NC9	NC32	AY26
H19	NC10	NC33	AY28
H27	NC11	NC34	AY30
H29	NC12	NC35	AY32
H31	NC13	NC36	AY25
AB7	MIPI_DSI_LDO	NC37	AY29
AJ16	NC15	NC38	BA24
AJ7	NC16	NC39	BA30
AP2	NC17	NC40	BB29
AT19	NC18	NC41	BC30
AU16	NC19	NC42	BD1
AU24	NC20	NC43	BD29
AV21	NC21	NC44	BF2
AV23	NC22	NC45	BF40
AV25	NC23		

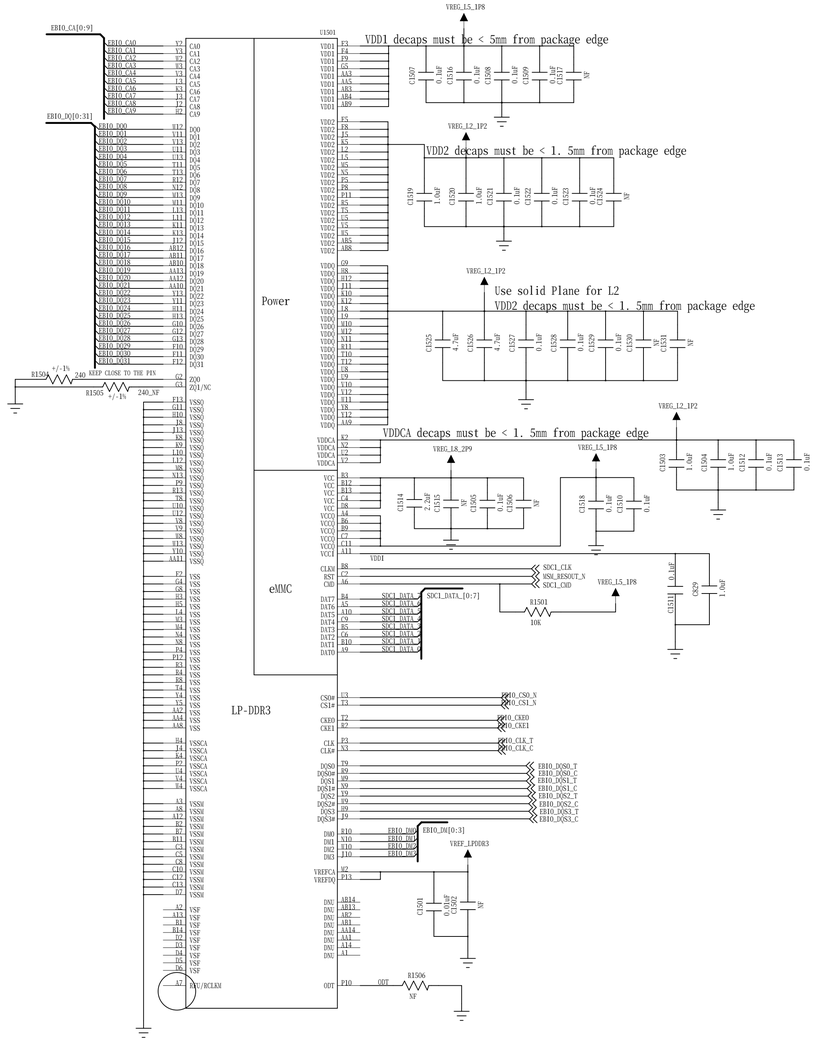
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

	MICRO/HYNIX	SAMSUNG
U1501.G3	ZQ1	NC
U1501.A7	RFU	RCLK



Note: Pull-up resistors on SDC1_DATA are PCB and eMMC vender dependent

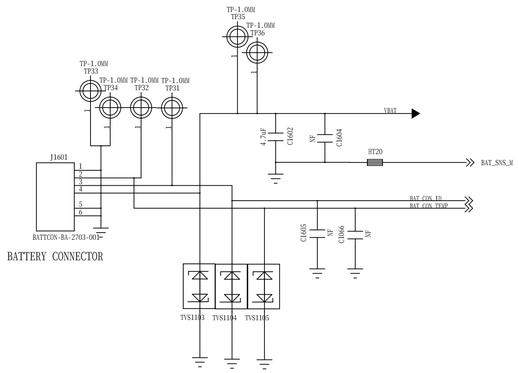
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

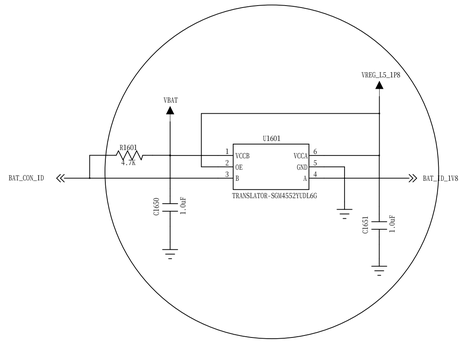
Title

Size | Sheet base | Rev

Date: | Sheet of



BATTERY CONNECTOR



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

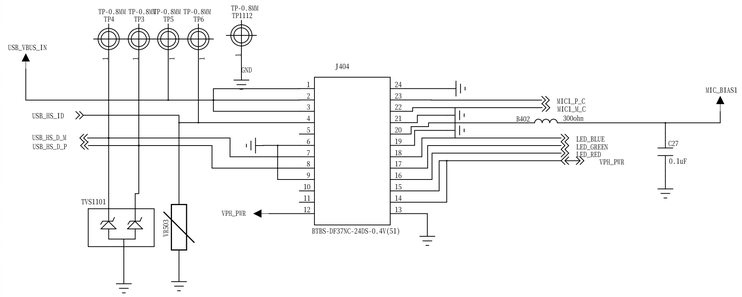
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

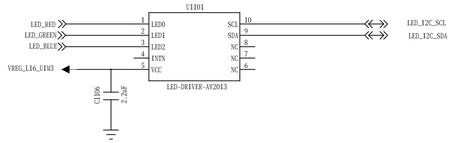
Size | Sheet Name | Rev

Date: | Sheet of

Sub Board Connector



MAIN MIC+USB+RGB



RGB

I2C Address	
Write	8B0x8A
Read	8B0x8B

Power on sequence: can be applied in any order

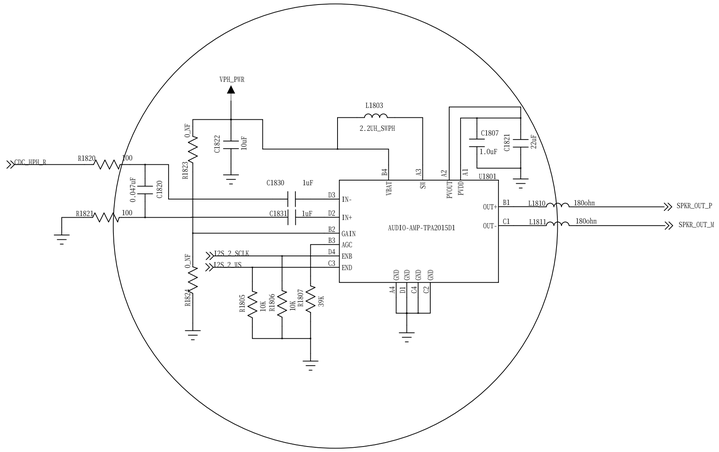
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

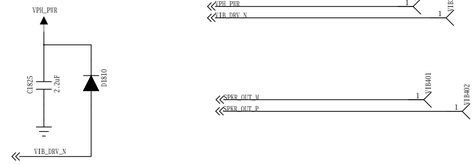
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

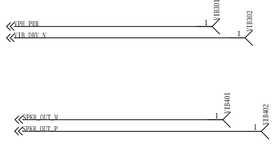
Title

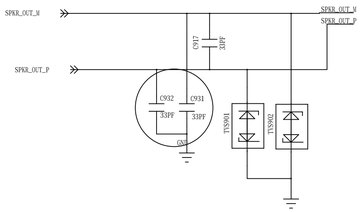
Size | Sheet Name | Rev

Date: | Sheet of

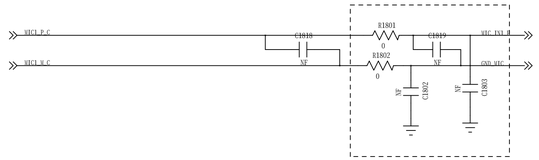




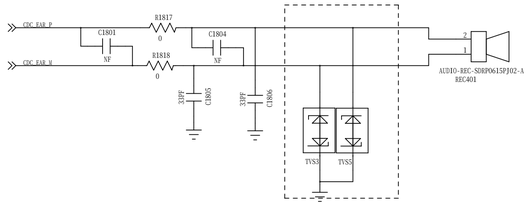




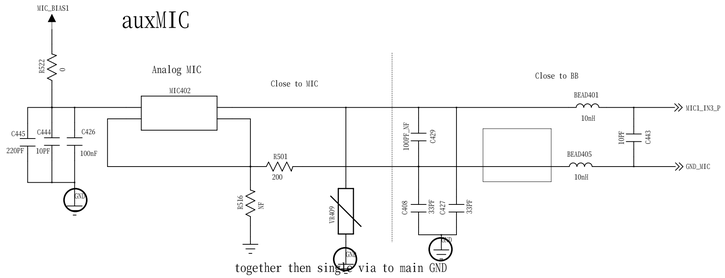
PRIMARY MIC



RECEIVER



auxMIC



together then single via to main GND

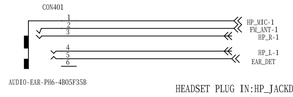
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

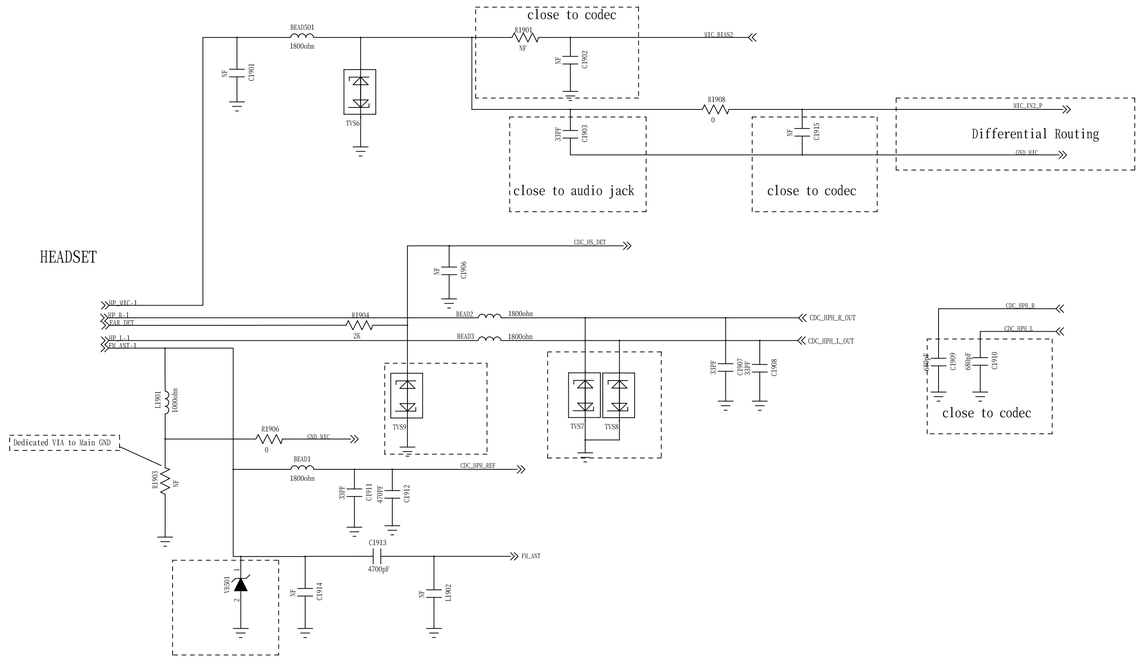
Title

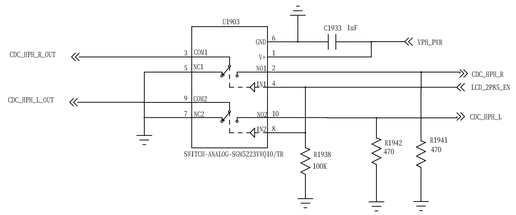
Size | Sheet Name | Rev

Date: | Sheet of



HEADSET PLUG IN:HP_JACKDET "0"
HEADSET PLUG OUT:HP_JACKDET "1"





IN=0, COM Connected to NC;
IN=1, COM Connected to NO;

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

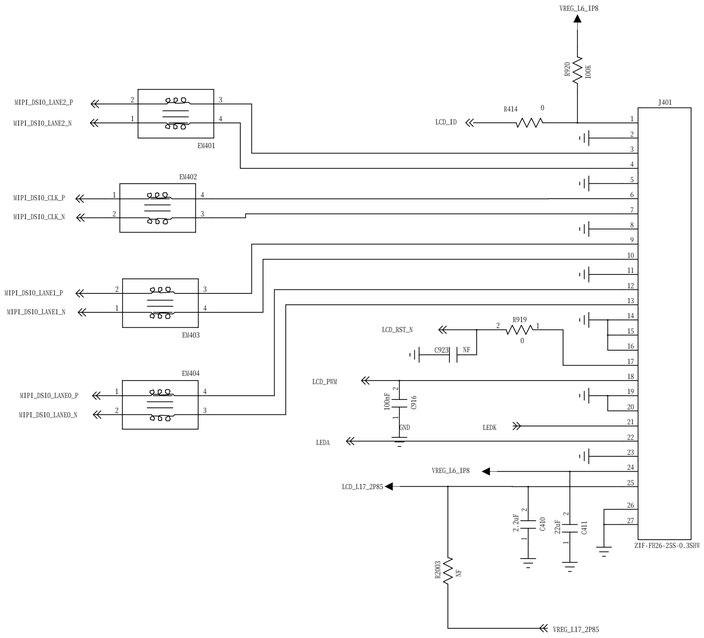
LCD

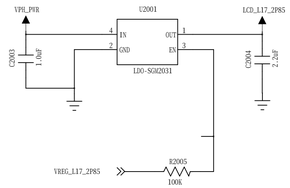
5.0' FWVGA

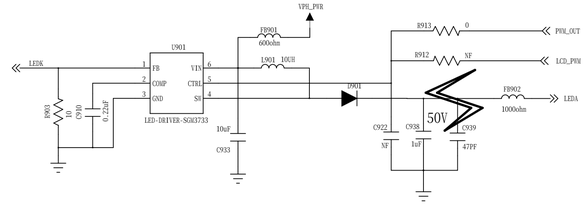
Module: KTF-050-D54980F

Driver: OTM8019A

Power on sequence: L6(IOVCC)/L17(VCC) can be applied in any order







Backlight LED Driver

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

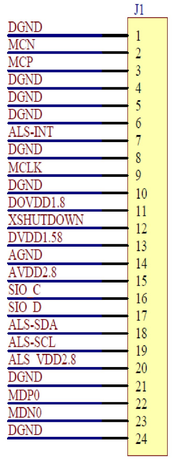
Title

Size | Sheet Name | Rev

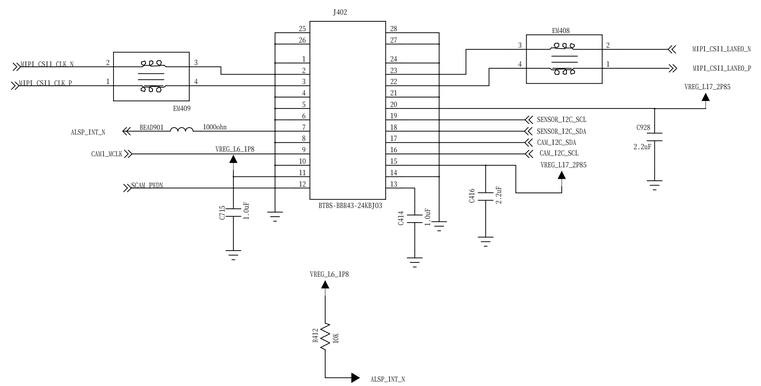
Date: | Sheet of

Sub Camera

FRONT_camera+LP Sensor+SPK



BBR13-24K6J01



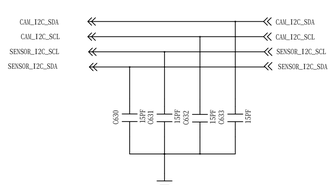
Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible

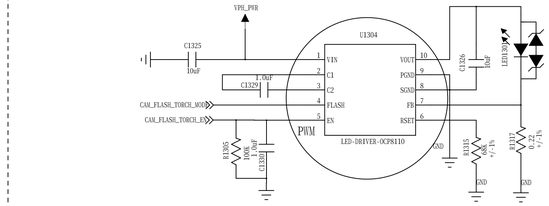
light sensor 0x90 Write(STK3171)

light sensor 0x46 Write (LTR-558ALS)

Front Camera: 2M Module: Sensor: OV2680	I2C Address	
	Write	0x20
	Read	0x21

Power on sequence: L6(DOVDD)/L17(AVDD) can be applied in any order





OCP8111 R1315=68K R1317=0.22

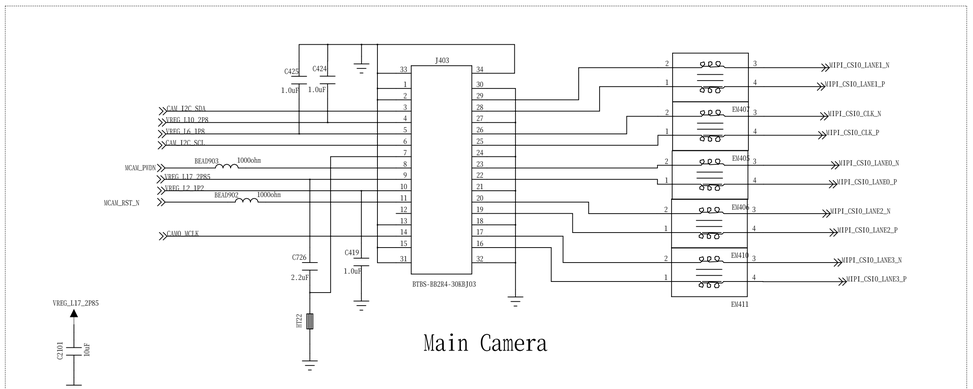
OCP8110 R1315=82K R1317=0.47

FLASH LED

light sensor 0x90 Write(STK3171)

light sensor 0x46 Write (LTR-558ALS)

Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible



Main Camera

Rear Camera: 8Mp Module: Sensor: OV8865	I2C Address	
	Write	0x6C
	Read	0x6D

Power on sequence: L6(DOVDD) must before L17(AVDD)

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

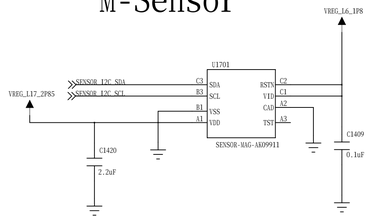
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

M-Sensor



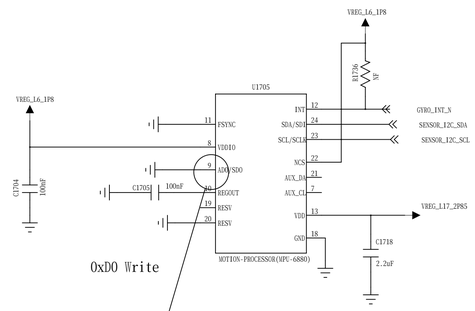
AKM: AKO9911

I2C Address@CAD=0	
Write	0x18
Read	0x19

I2C Address@CAD=1	
Write	0x1A
Read	0x1B

Power on sequence: L6/L17 can be applied in any order

Gyroscope



AD0=0 7b1101000 0x68

AD0=1 7b1101001 0x69

InvenSense: MPU6880

I2C Address@AD0=0	
Write	0xD0
Read	0xD1

I2C Address@AD0=1	
Write	0xD2
Read	0xD3

Power on sequence: L6/L17 can be applied in any order

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

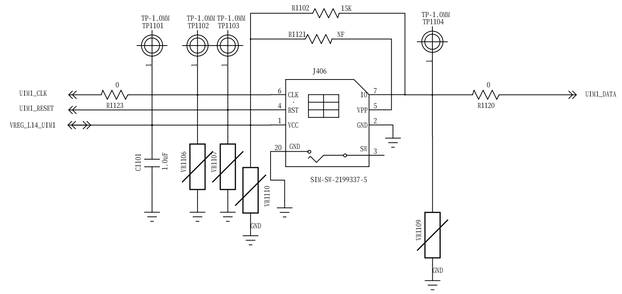
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

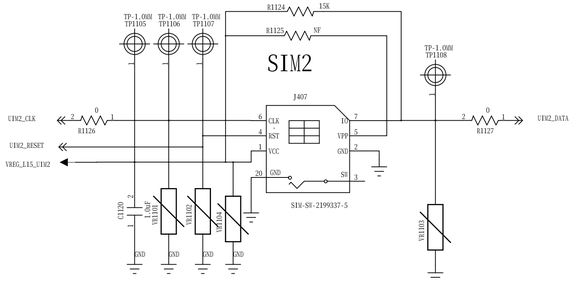
Size | Sheet Name | Rev

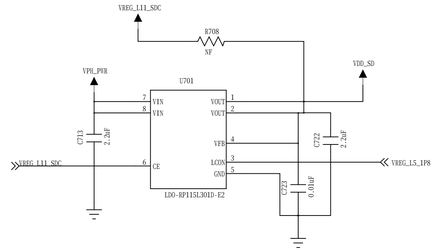
Date: | Sheet of

SIM1

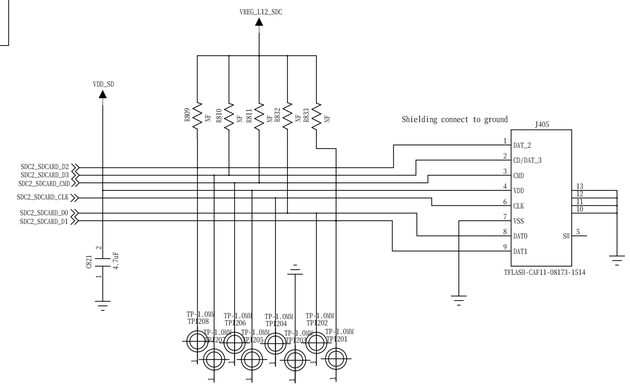


SIM2





SD CARD



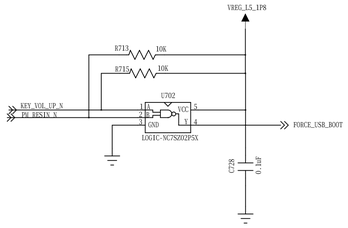
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

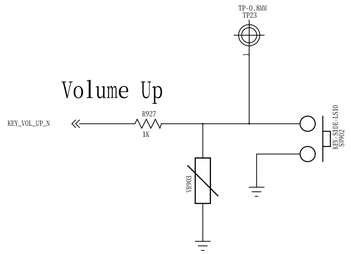
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

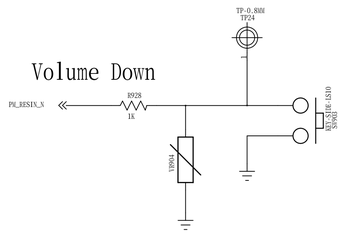
Title

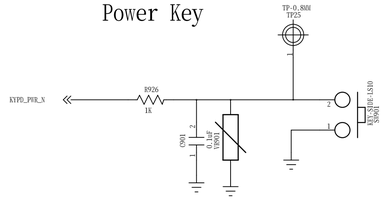
Size | Sheet base | Rev

Date: | Sheet of









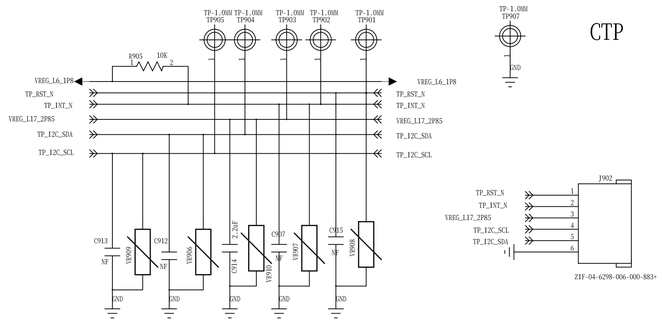
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



CTP

Power on sequence: Single Supply

FocalTech: FT5336	
I2C Address	
Write	0x70 It can be changed by FACTORY.
Read	0x71

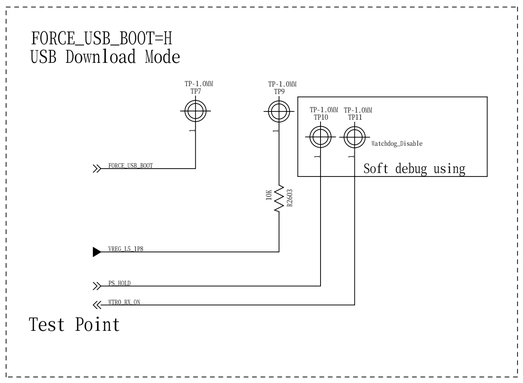
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

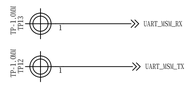
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

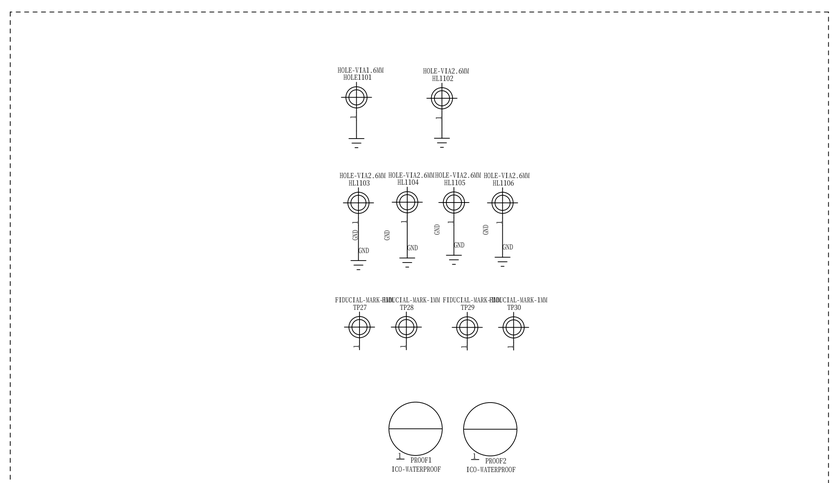
Size | Sheet Name | Rev

Date: | Sheet of





JTAG DEBUG



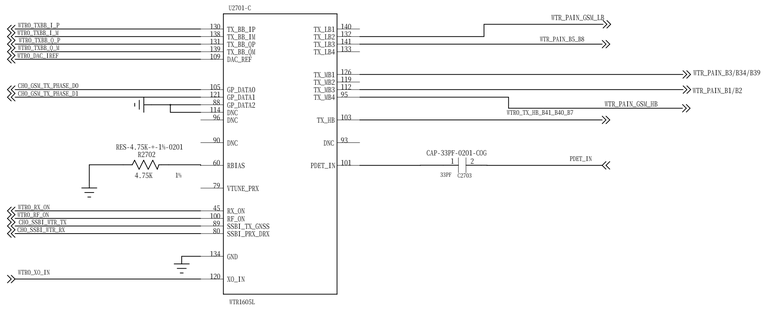
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



use C2701 for both SPDT power supply, and use C2705 for both SPDT control line

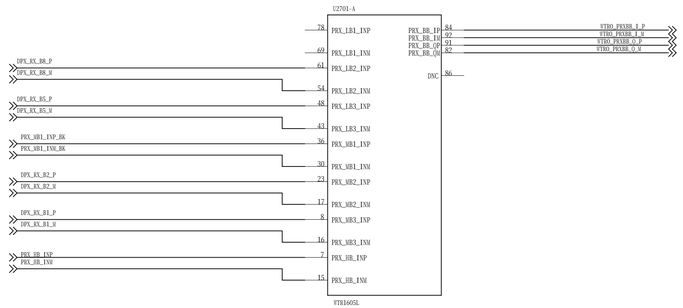
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

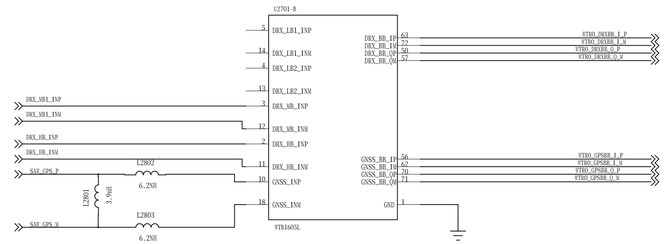
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of





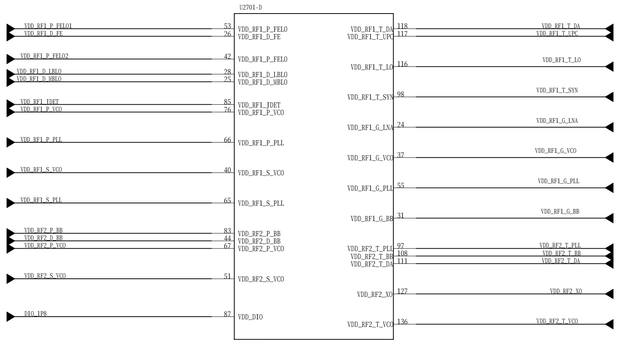
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

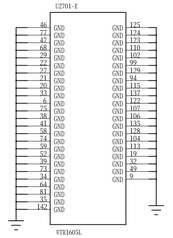
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



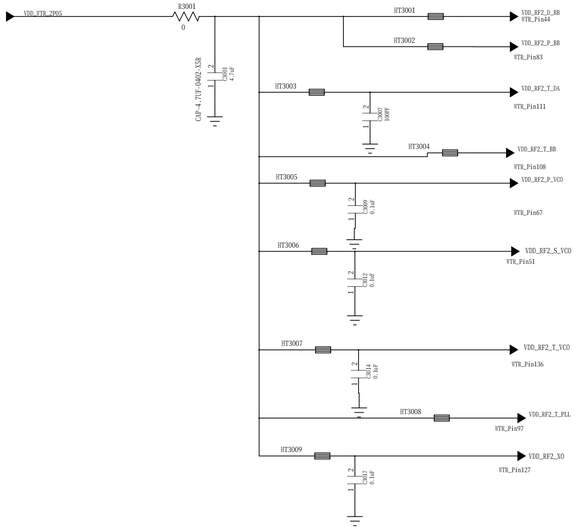


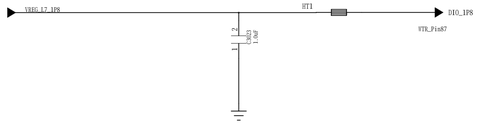
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

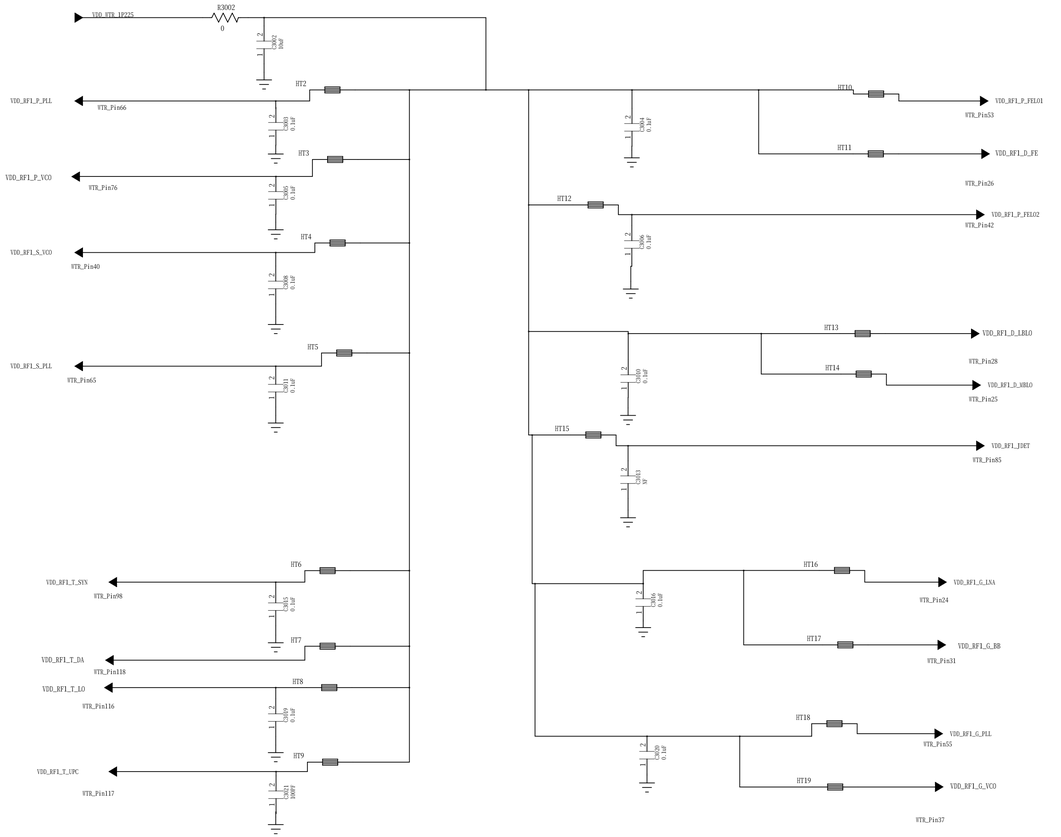
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	







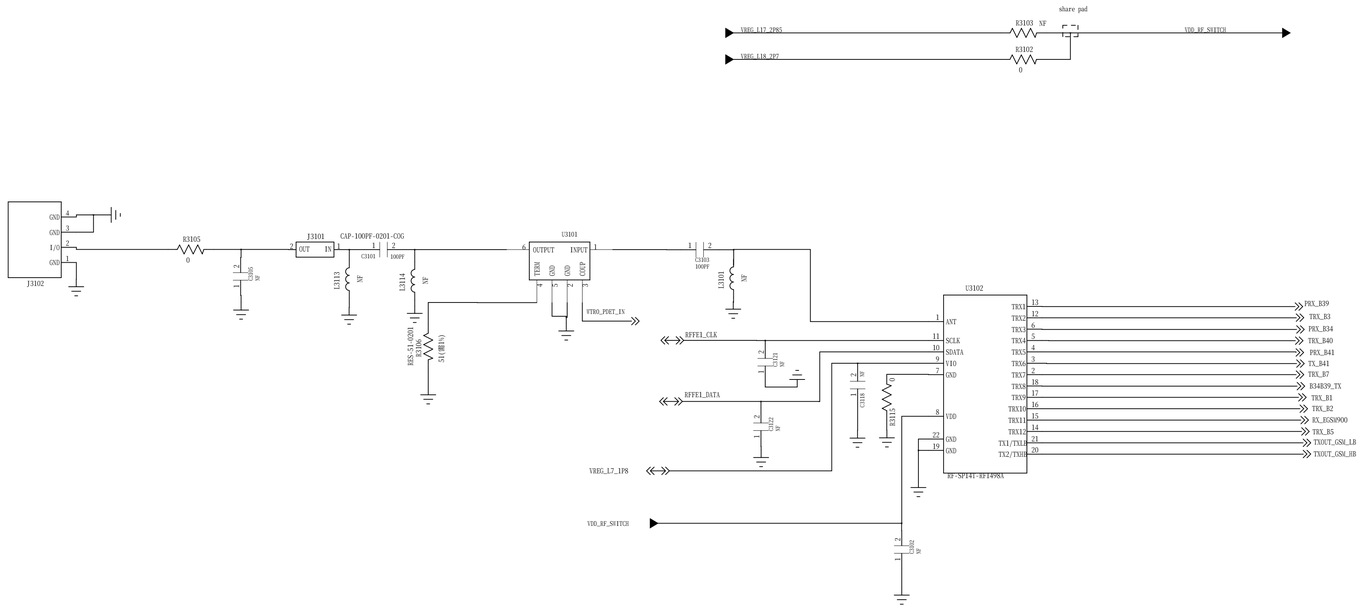
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet base | Rev

Date: | Sheet of



	ANT_SWITCH_CTL0	ANT_SWITCH_CTL1
TRX_B7		
TRX_B40		
QFE2320_ANT		
TRX_B38_41		

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

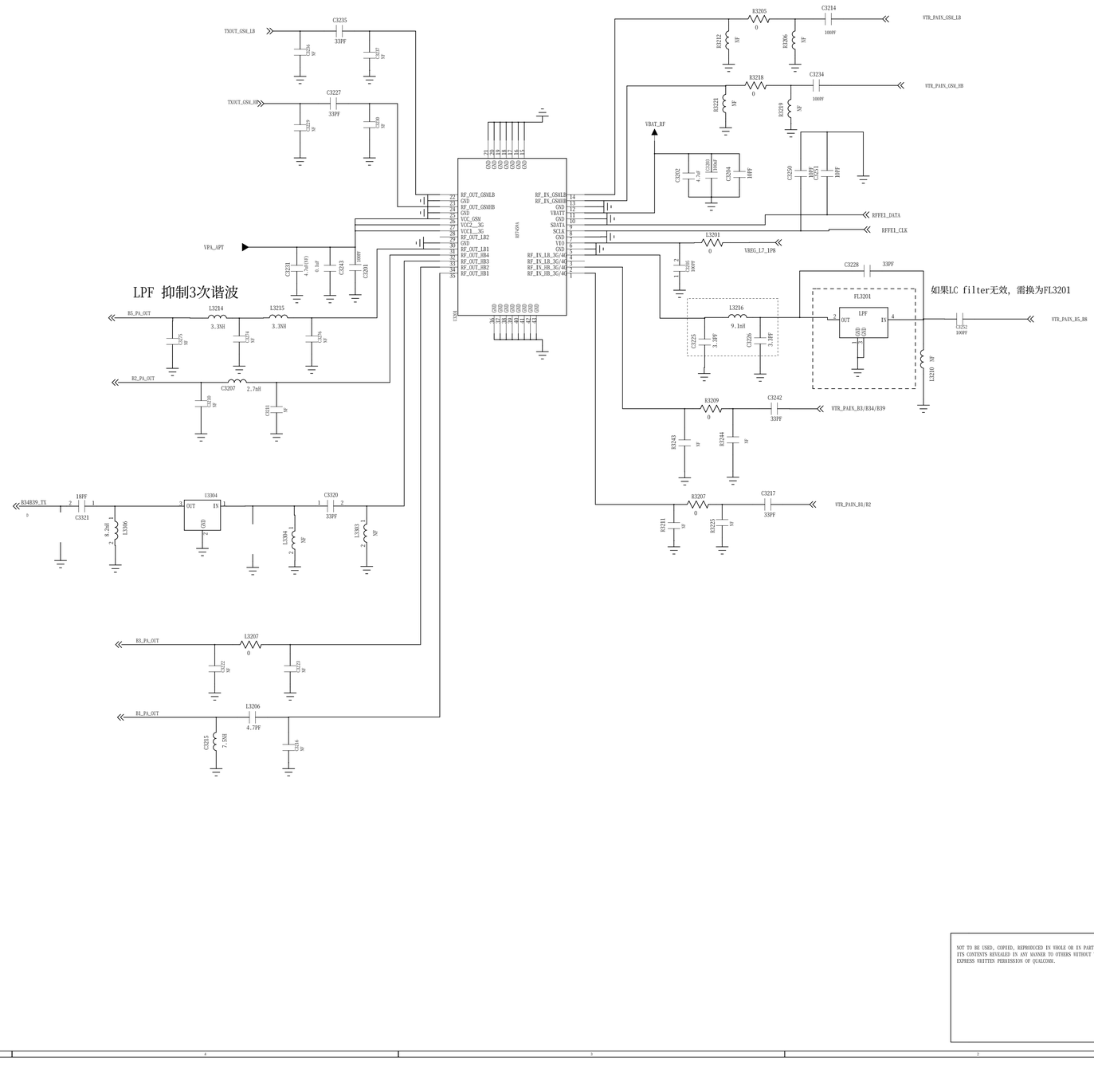
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

MMMB PA : B1,2,3,5,17 2G HB, 2G LB



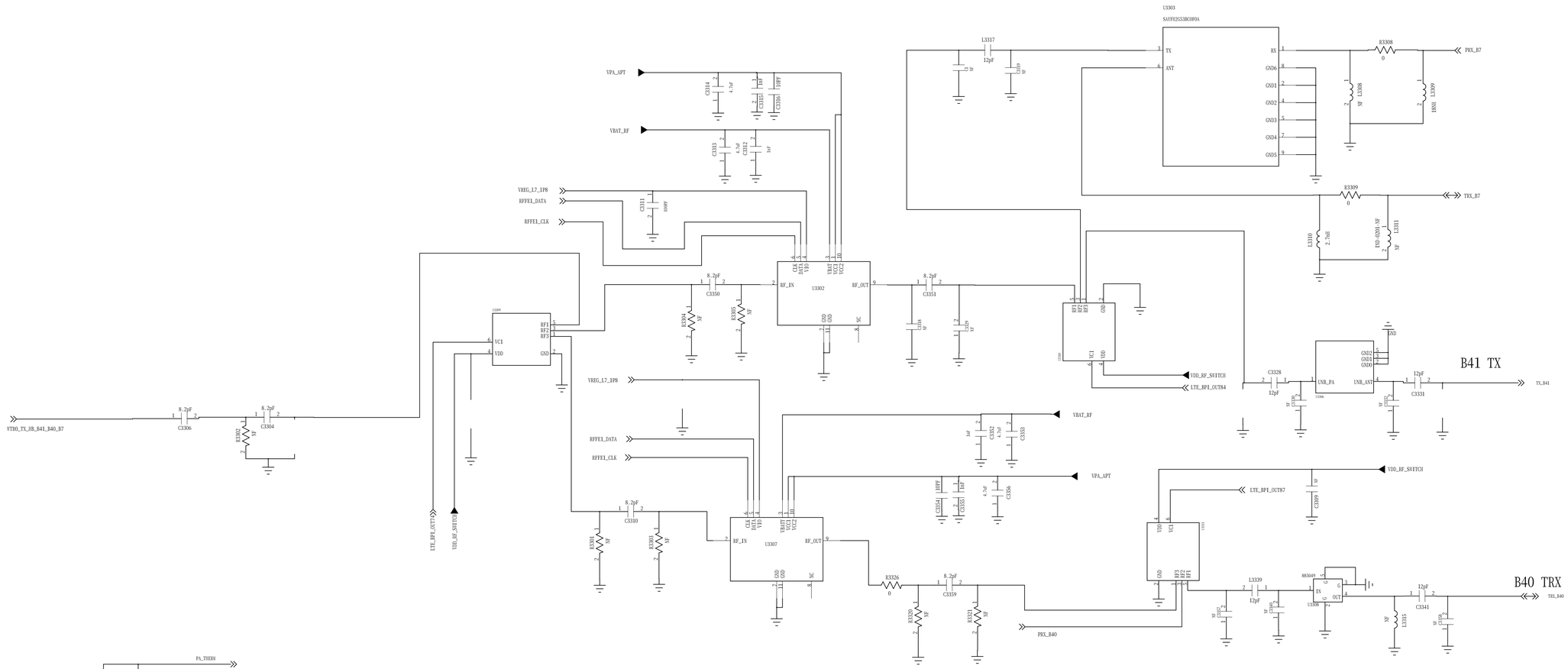
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

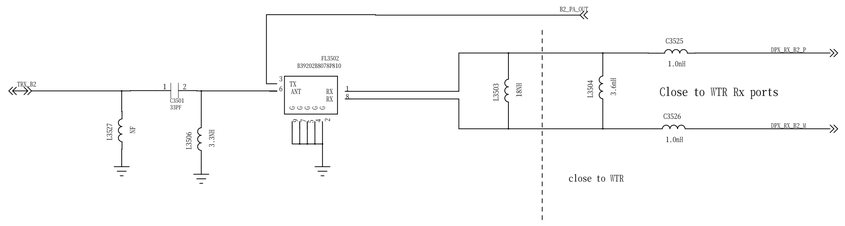
Date: | Sheet of

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

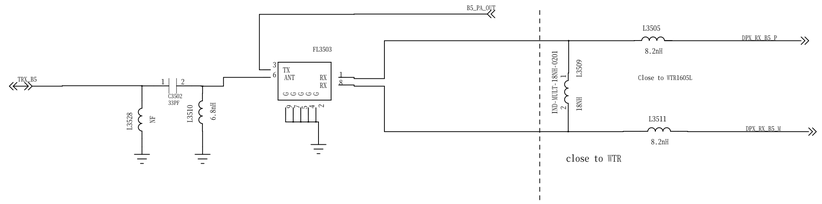
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

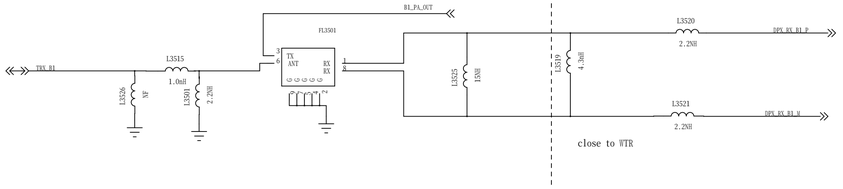
Size	Sheet Name	Rev
Date:	Sheet of	



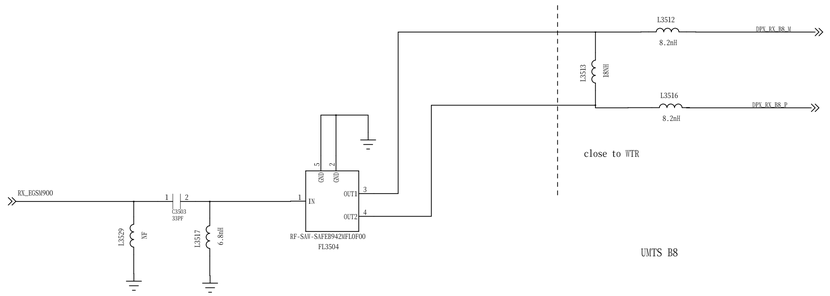
UMTS B2



UMTS B5



UMTS B1



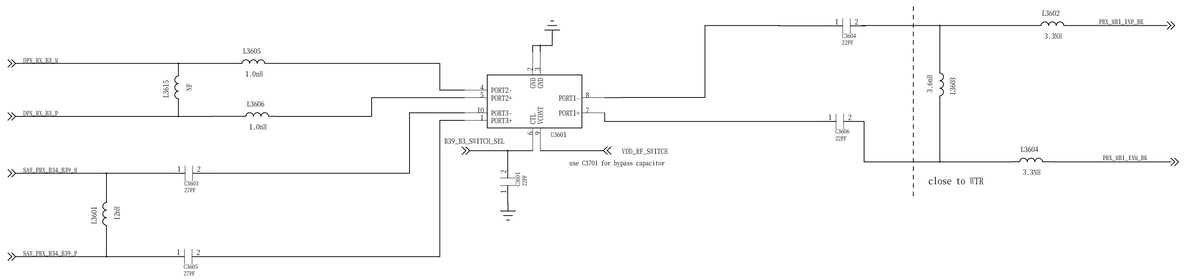
UMTS B8

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

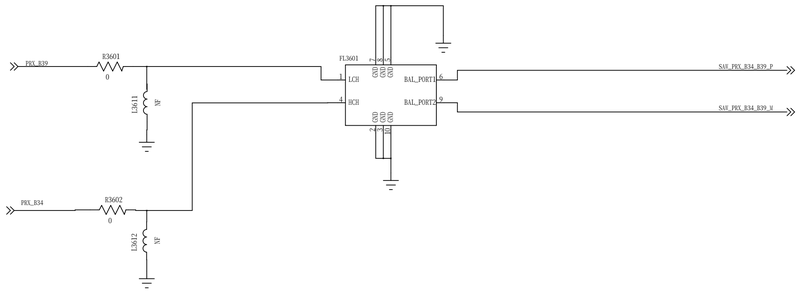
Title

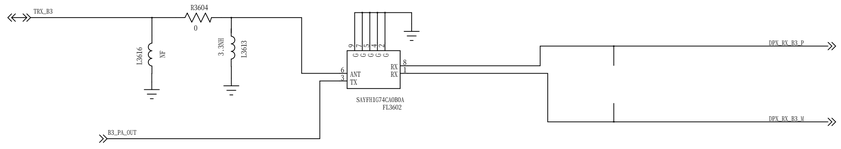
Size	Sheet Name	Rev
Date:	Sheet of	



B34,B39/B3 RX Switch

MODE	B39_B3_SWITCH_SEL
PORT1 TO PORT2	L
PORT1 TO PORT3	H





UMTS B3

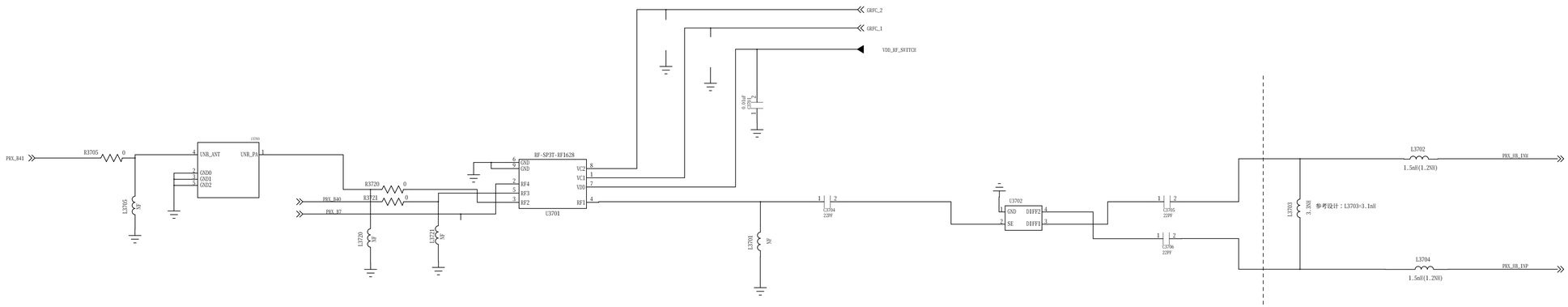
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



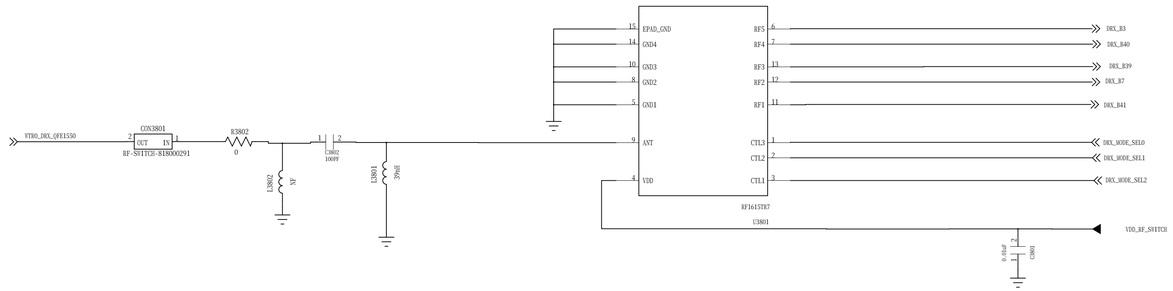
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of



RX BAND SELECT LOGIC

	PORT	DRX_MODE_SEL0	DRX_MODE_SEL1	DRX_MODE_SEL2
DRX_B7	RF1			
DRX_B40	RF2			
DRX_B41	RF3			
DRX_B39_B1	RF4			
DRX_B3	RF6			

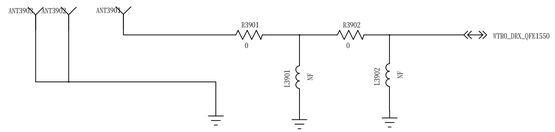
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of

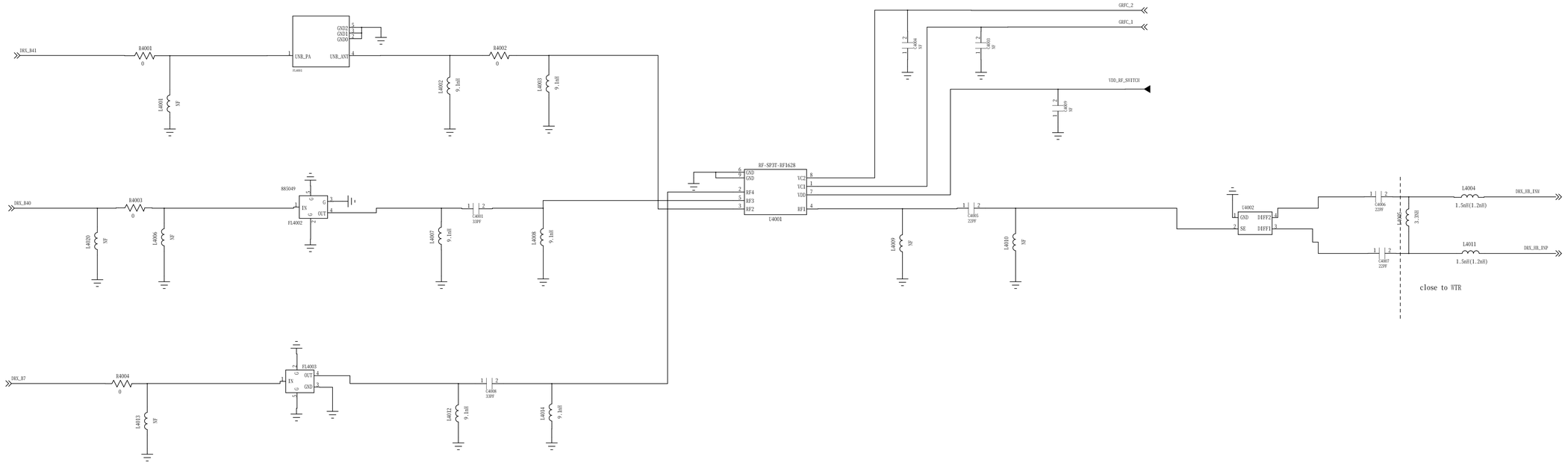


NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	



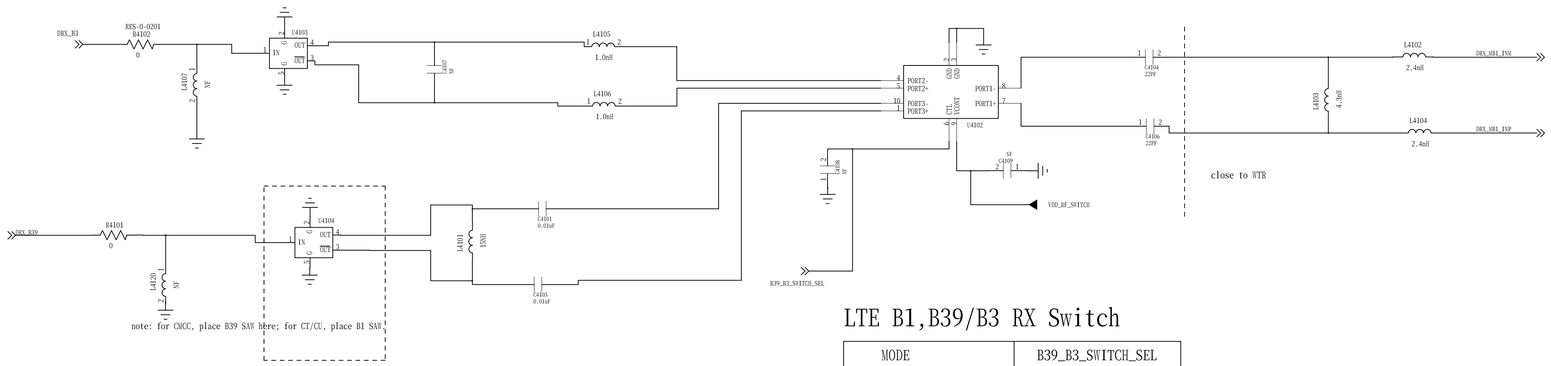
NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size | Sheet Name | Rev

Date: | Sheet of


DRX_B3
R4102
0
L4107
XF
L4103
L4105
1.0nH
L4106
1.0nH
C4107
XF
PORT2-
PORT2+
PORT3-
PORT3+
PORT1-
PORT1+
GND
GND
GND

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

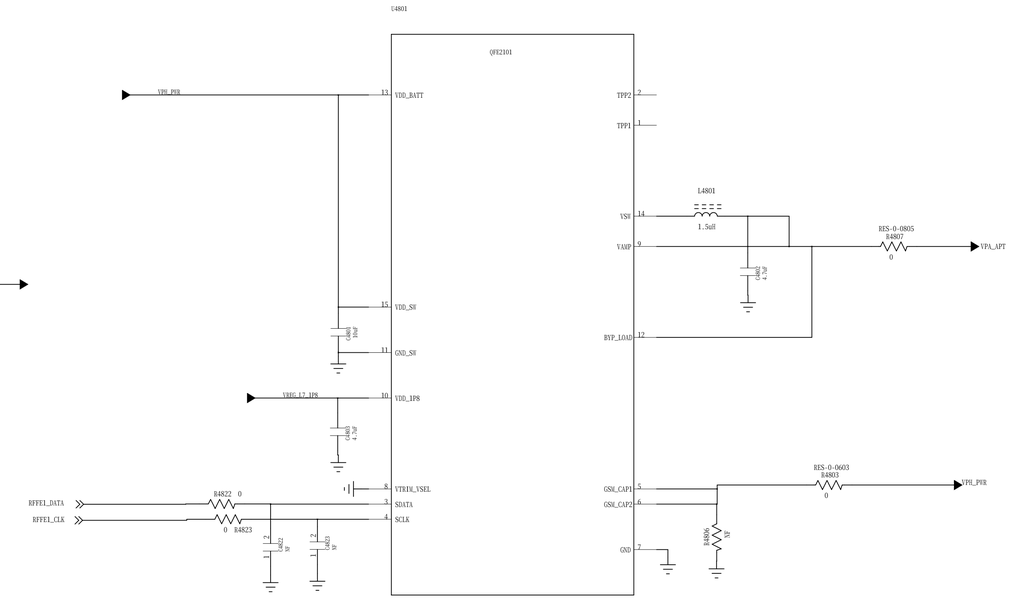
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev
Date:	Sheet of	

Note: Use a signle low impedance power plane/fill VPA for all ET PA VCC1/VCC2.

Refer to 80-NA681-91 rev.B or later revisions for more layout details.



NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

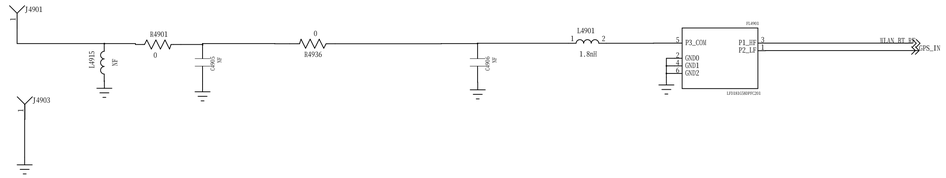
QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

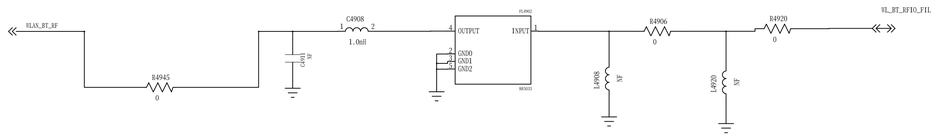
Title

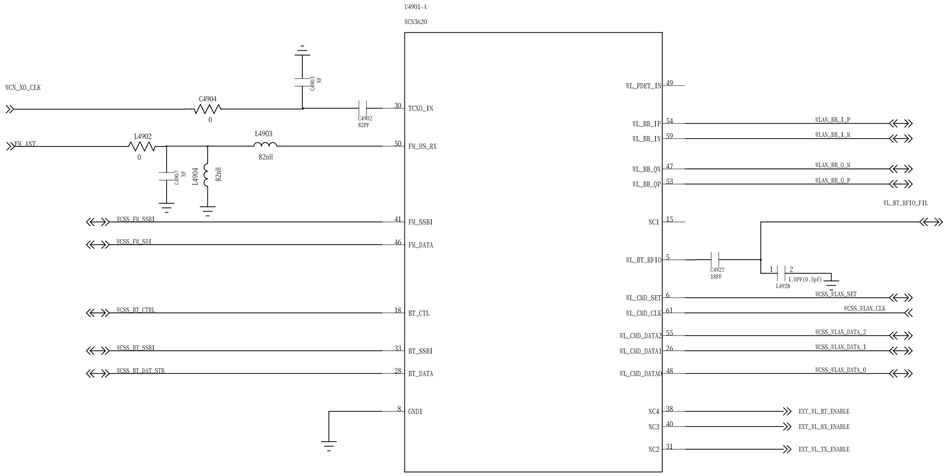
Size | Sheet Name | Rev

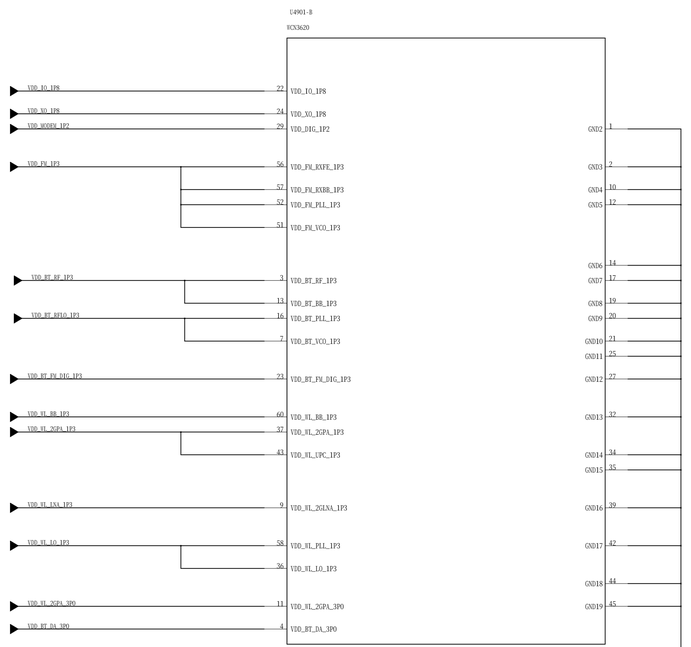
Date: | Sheet of

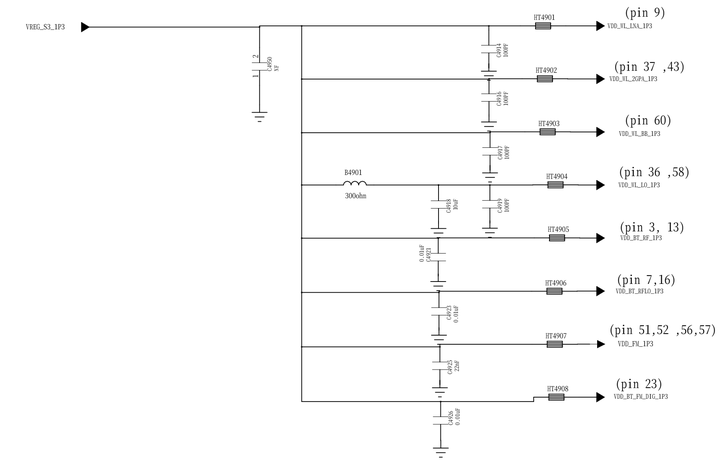
WIFI/BT

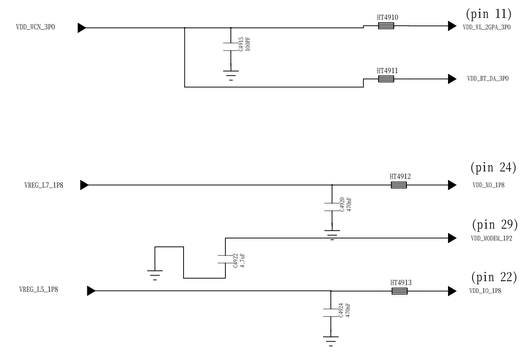












NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
SCHEMATIC, CCA, MAIN BOARD, QRD8916,2-4-2, CHCC CSFB, EVT

Size	Sheet name	Rev
Date:	Sheet of	

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated

5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

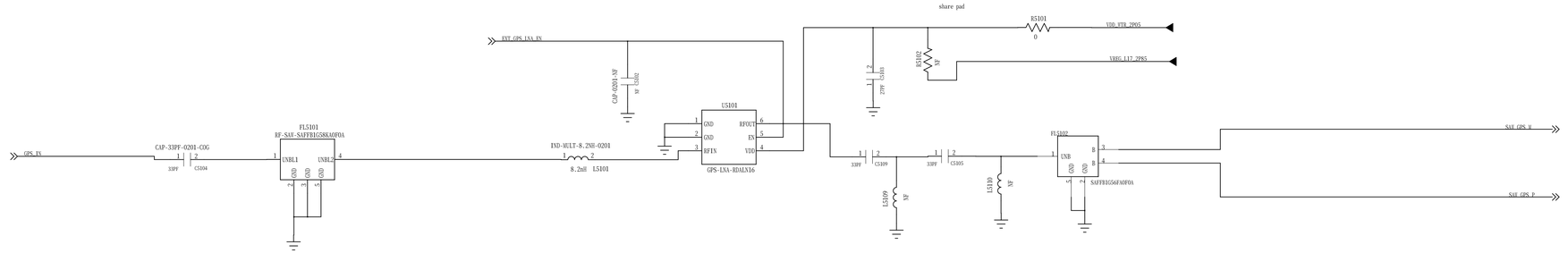
Title

SCHEMATIC, CCA, MAIN BOARD, QRD8916,2-4-2, CHCC CSFB, EVT

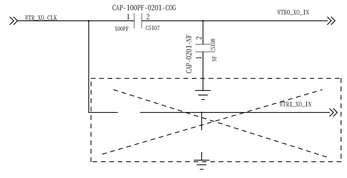
Size	Sheet Name	Rev

Date: | Sheet of

GPS



XO DISTRIBUTION



NC

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title
SCHEMATIC, CCA, MAIN BOARD, QRD8916_2-4-2, CHCC CSFB, EVT

Size | Sheet Name | Rev

Date: | Sheet of

NOT TO BE USED, COPIED, REPRODUCED IN WHOLE OR IN PART, NOR
ITS CONTENTS REVEALED IN ANY MANNER TO OTHERS WITHOUT THE
EXPRESS WRITTEN PERMISSION OF QUALCOMM.

QUALCOMM Incorporated
5775 Morehouse Drive
San Diego, CA 92121-1714
U.S.A.

Title

Size	Sheet Name	Rev

Date: Sheet of